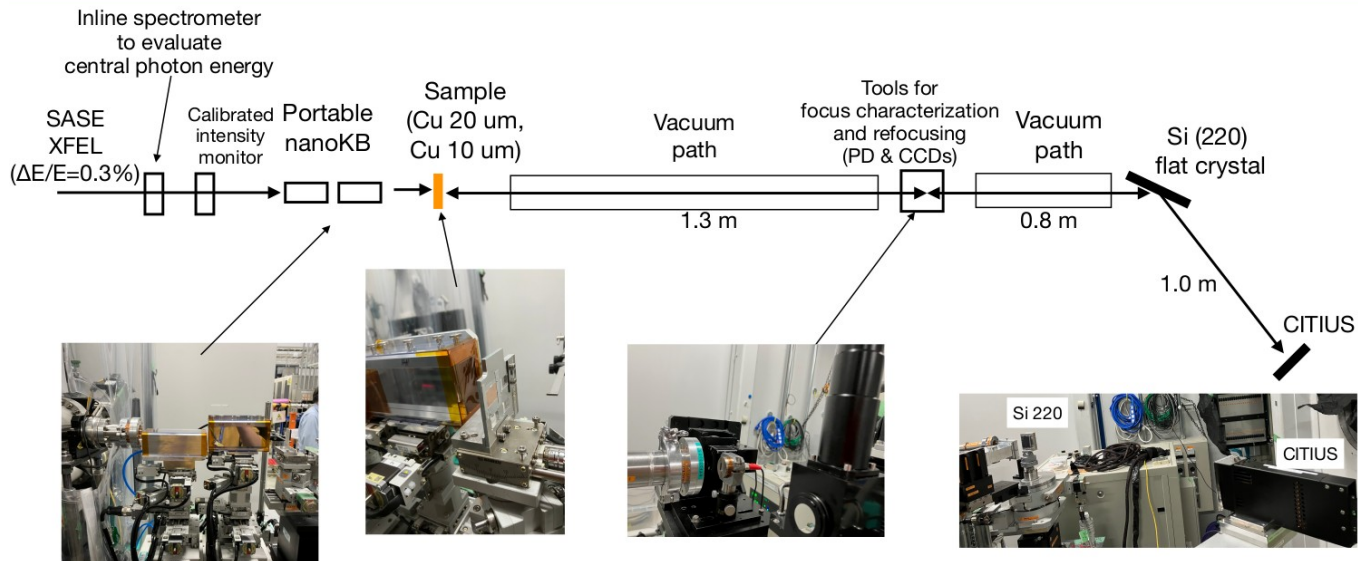


Pumping below K-edge with radiation resonant to K- α transition:

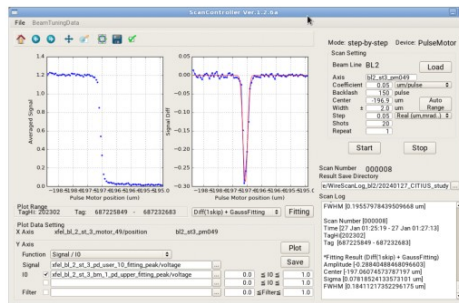
- Saturable absorption – measurements done by Ichiro
 - 26.01.2024-29.01.2024
 - 07.2024
- Modeling the RSA
 - XLO code for absorption
 - XATOM for occupations

Experimental data from Ichiro

Setup (SACLA BL2, EH3)



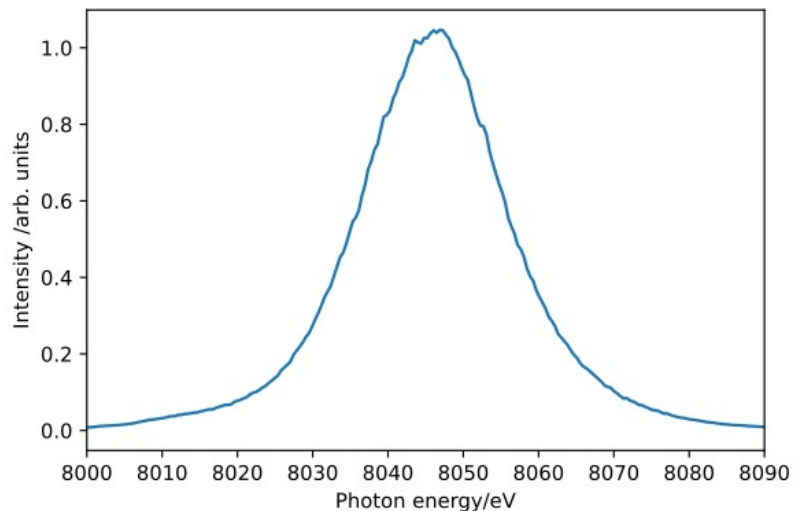
FWHM beam size@sample position: 180 nm (H) x 210 nm (V)



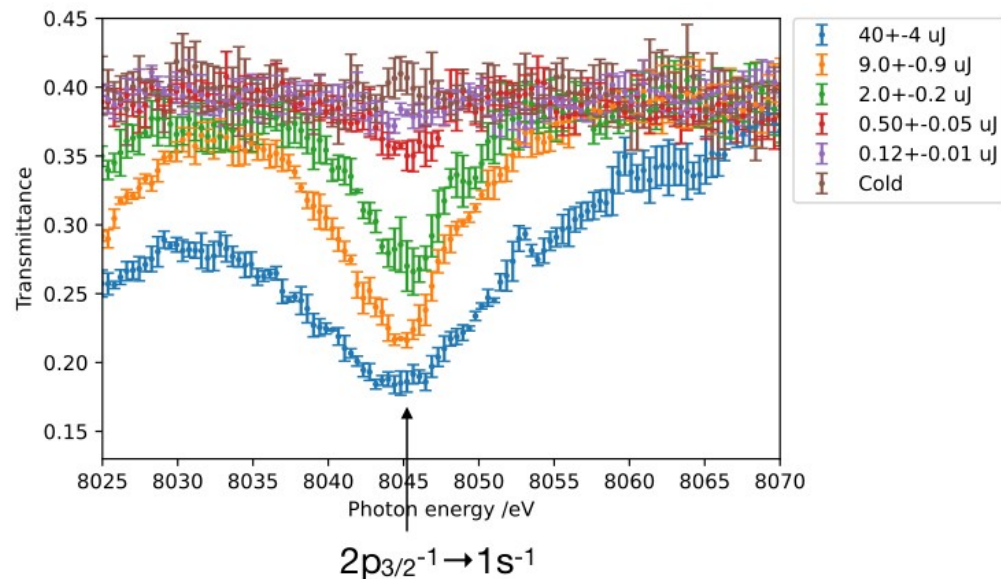
Experimental data from Ichiro

Central photon energy of XFEL beam: 8045 eV ~Ka1

Average spectrum of incident beam

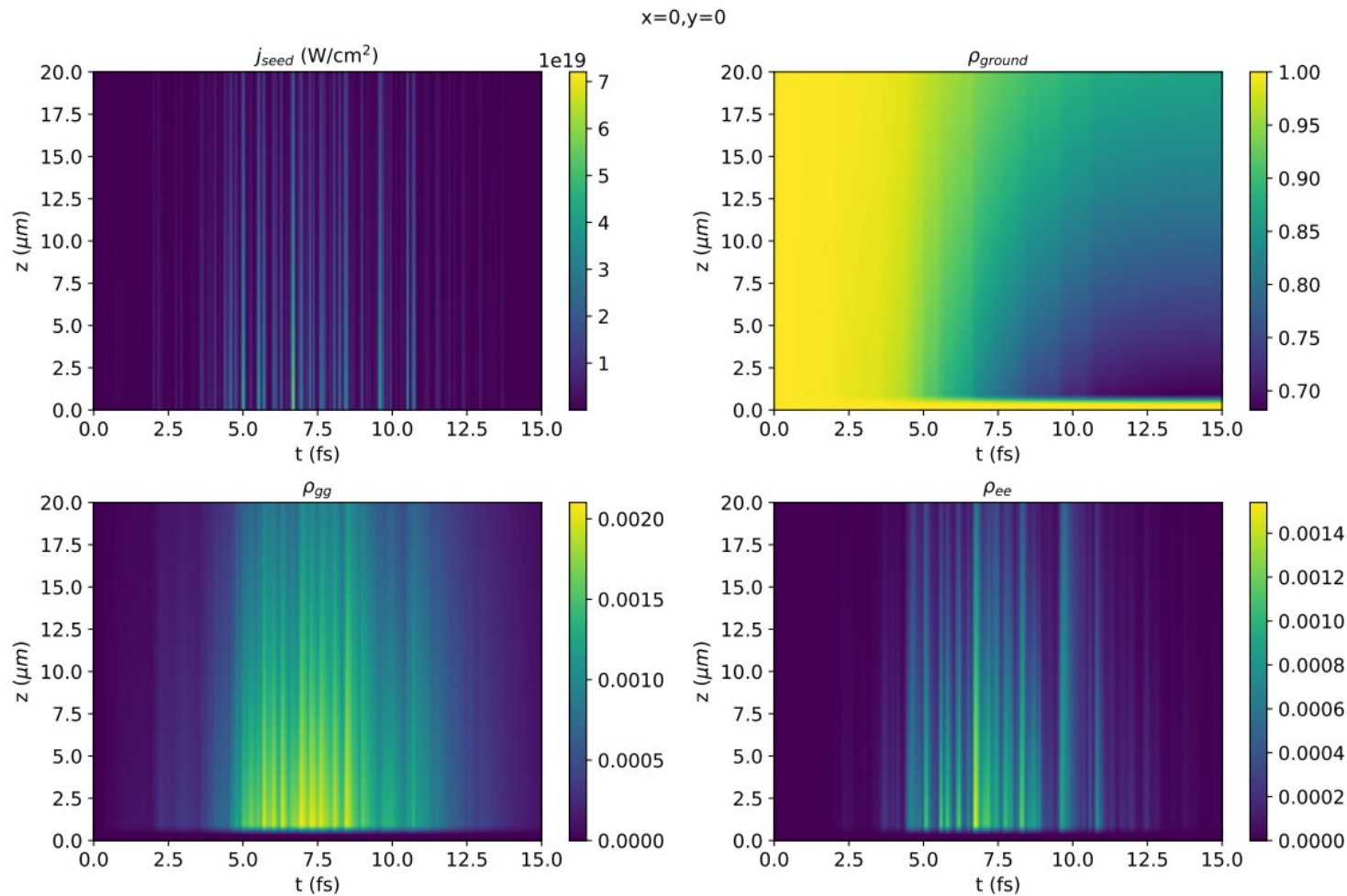


Transmittance (Cu 20 μ m)



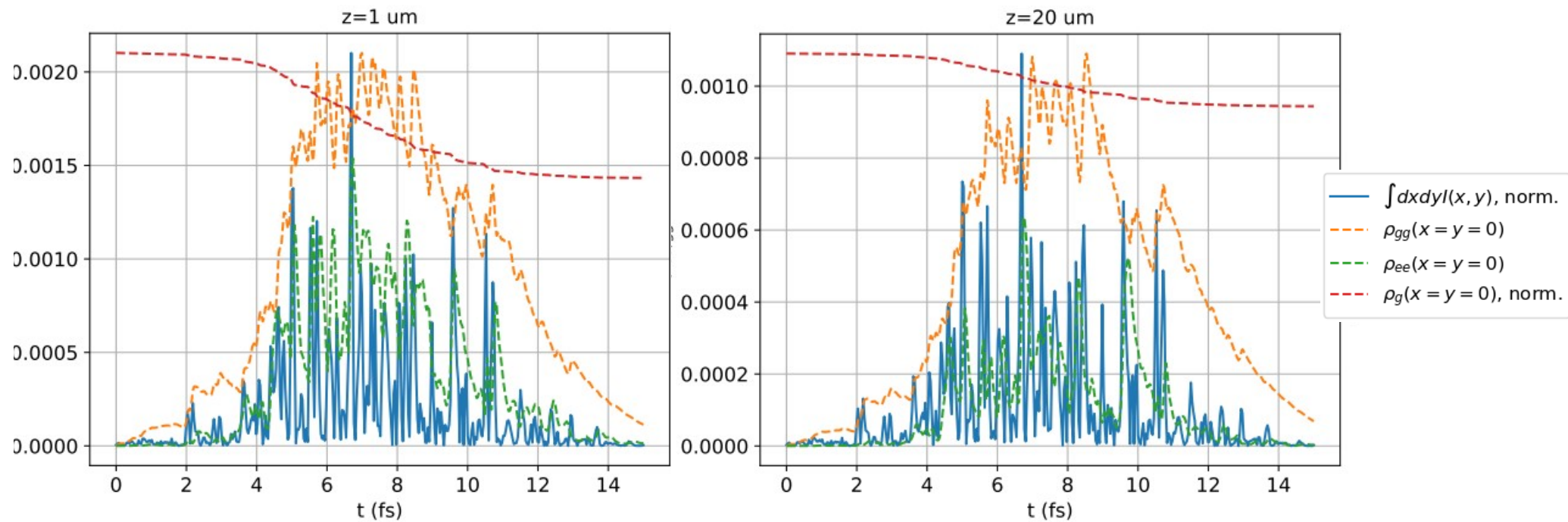
Behavior for a single run

40 uJ, FWHM: 180 nm x 220 nm, 6 fs

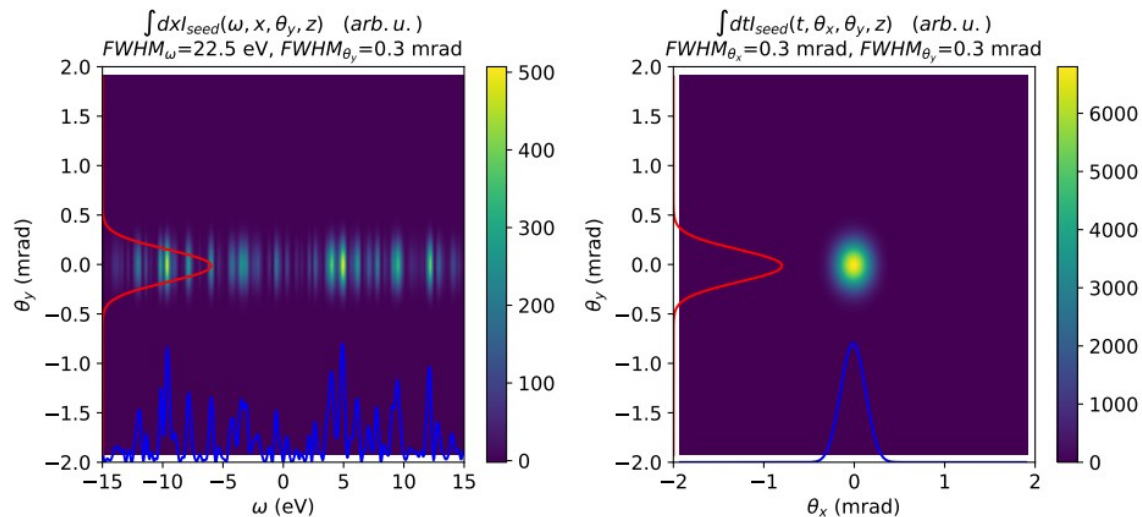


Behavior for a single run

40 uJ, FWHM: 180 nm x 220 nm, 6 fs



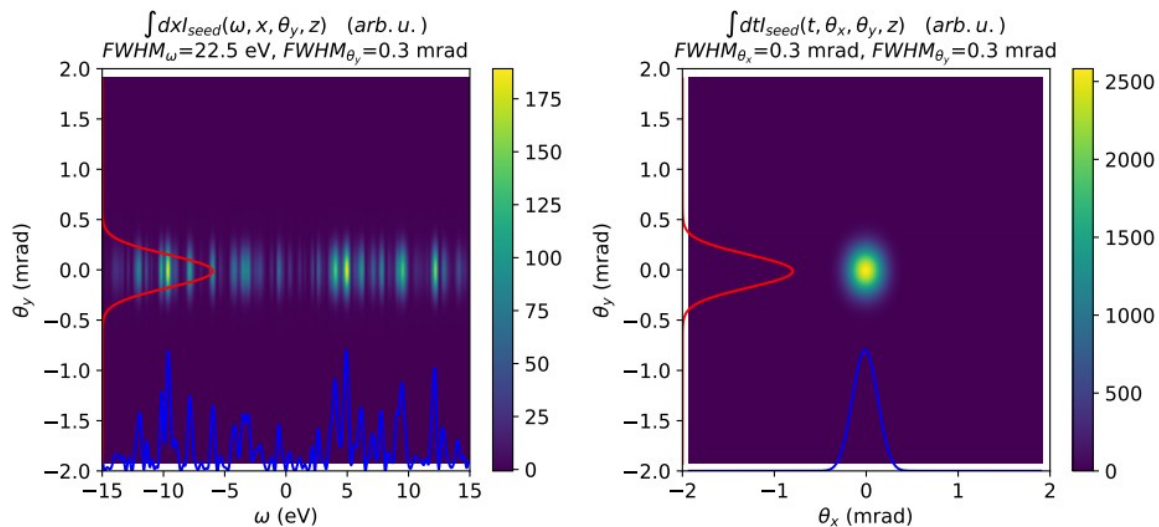
$z = 1 \text{ } \mu\text{m}$



Behavior for a single run

40 uJ,
FWHM: 180 nm x 220 nm, 6 fs

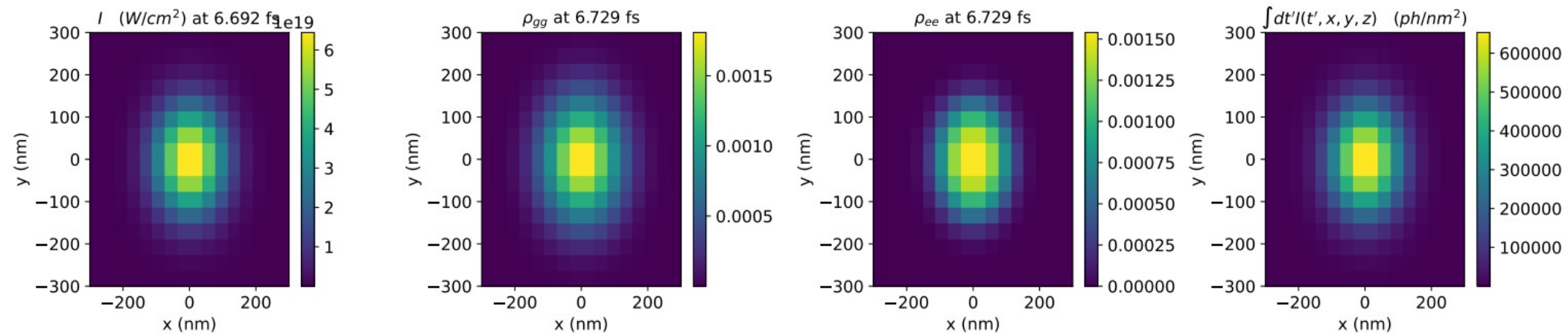
$z = 20 \text{ } \mu\text{m}$



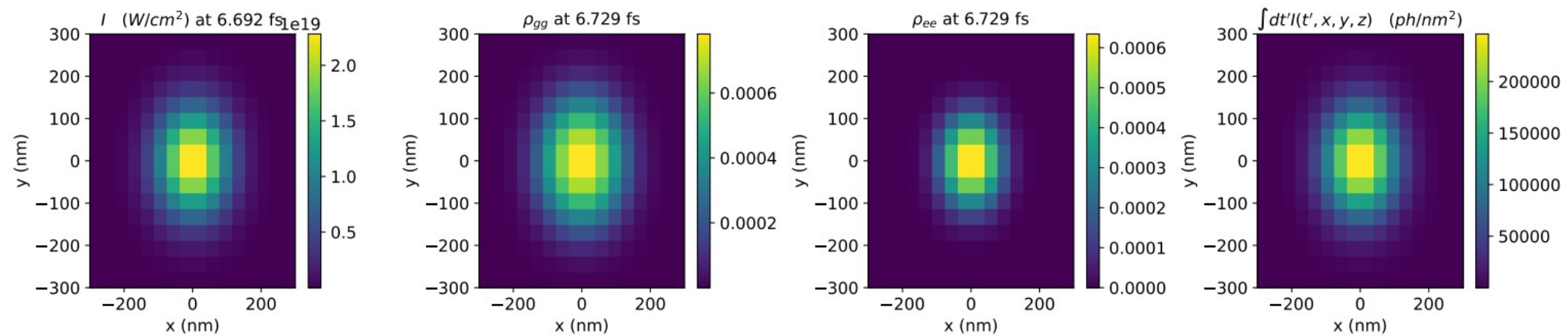
Behavior for a single run

40 μJ , FWHM: 180 nm x 220 nm, 6 fs

$z = 1 \mu\text{m}$, $N_{\text{seed}}^{\text{SF}} = 3.1\text{e}+10$

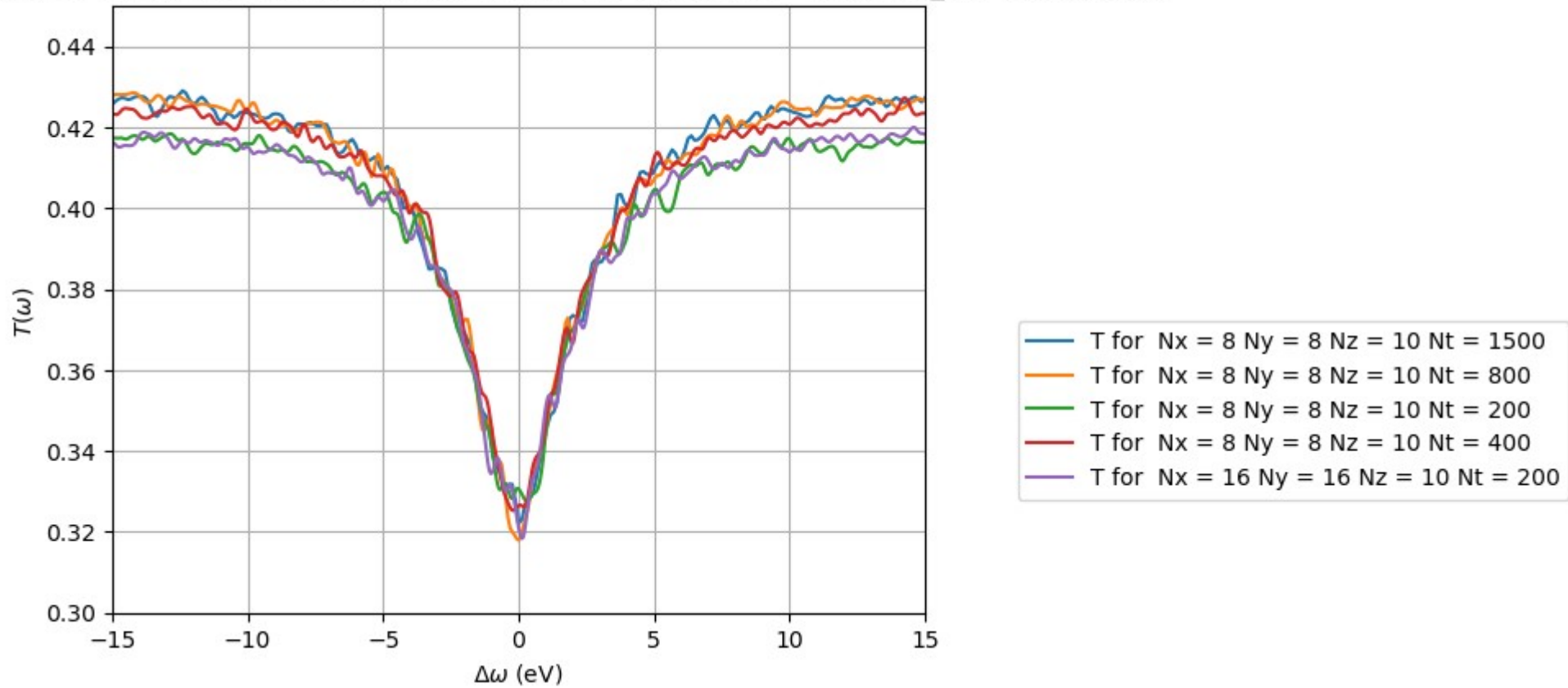


$z = 20 \mu\text{m}$, $N_{\text{seed}}^{\text{SF}} = 1.2\text{e}+10$



Sensitivity to grid (convergence)

average over 90 SASE reliaizations, convergence for $N_z=10$, $E_{\text{seed}}=40$ uJ, $\text{FEL_bw}=0.0015$ case

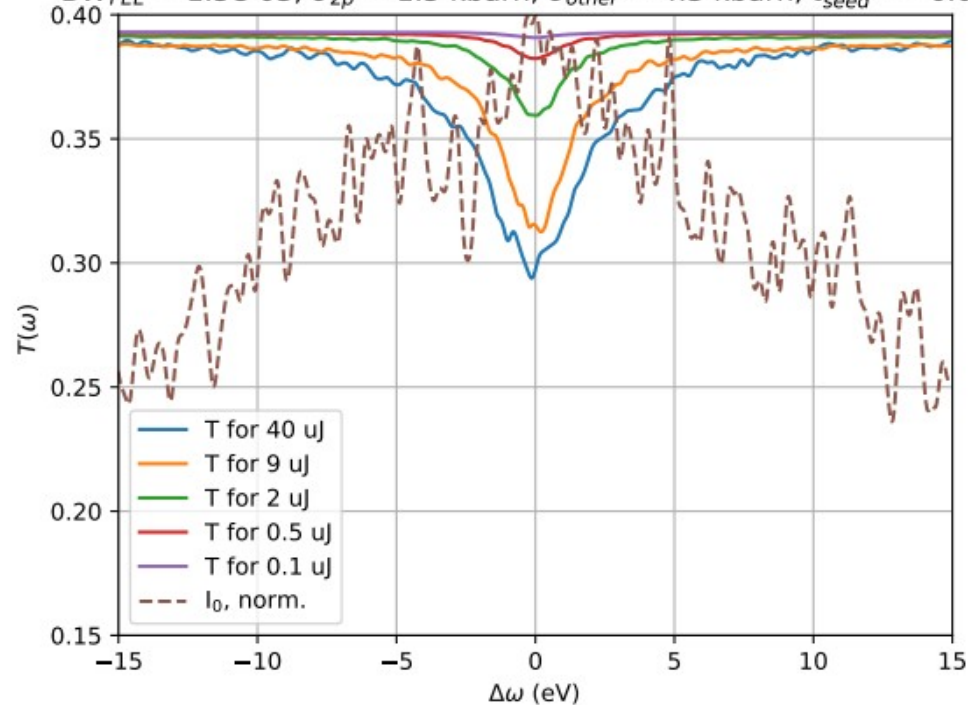


Sensitivity to FEL bandwidth

average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 200$

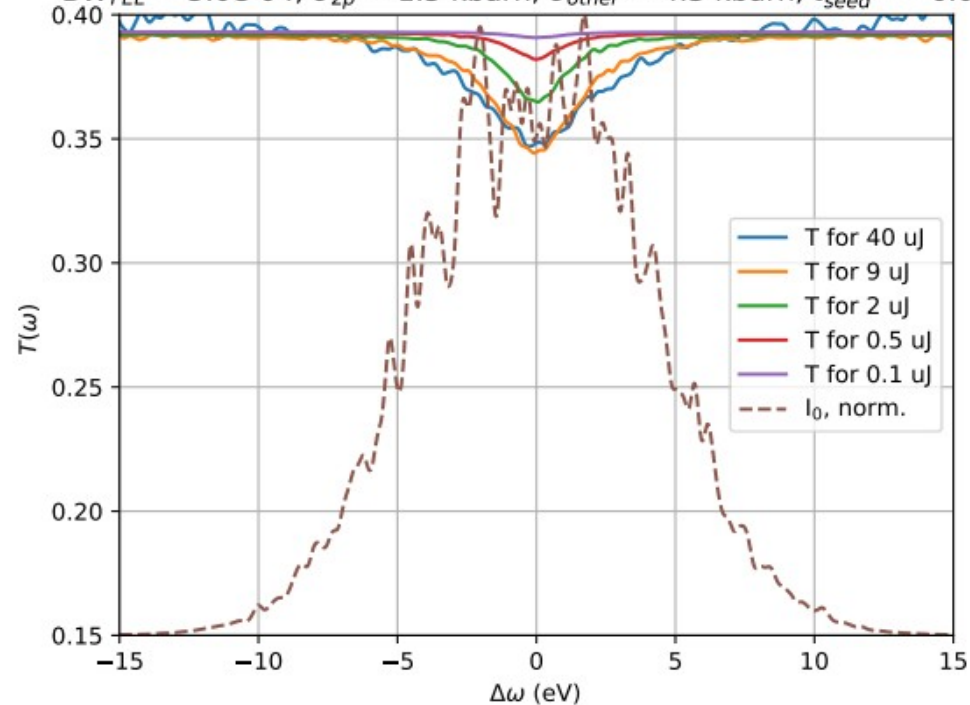
$BW_{FEL} = 1.5e-03, \sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 6.0 \text{ fs}$



average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 200$

$BW_{FEL} = 5.0e-04, \sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 6.0 \text{ fs}$

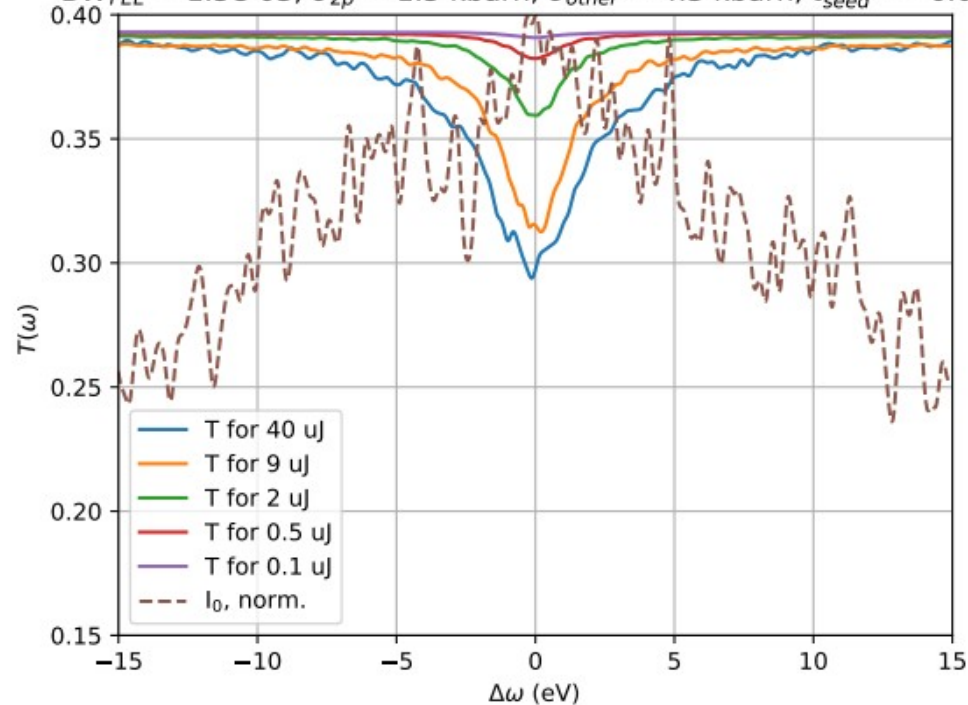


Sensitivity to absorption cross-section

average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 200$

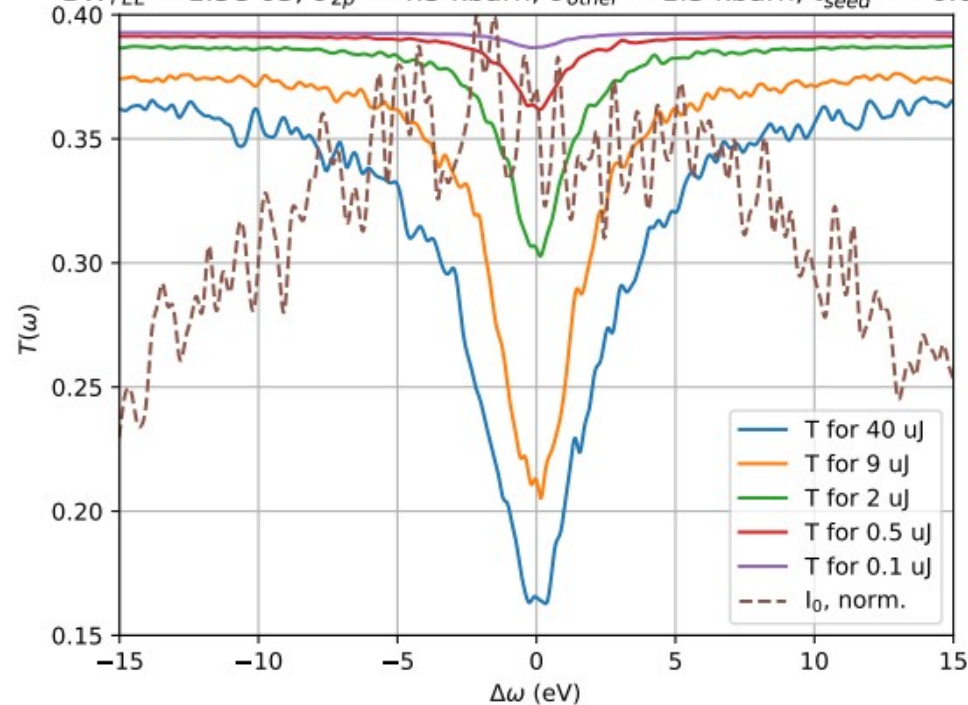
$BW_{FEL} = 1.5e-03, \sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 6.0 \text{ fs}$



average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 200$

$BW_{FEL} = 1.5e-03, \sigma_{2p} = 4.5 \text{ kbarn}, \sigma_{other} = 1.3 \text{ kbarn}, t_{seed}^{FWHM} = 6.0 \text{ fs}$

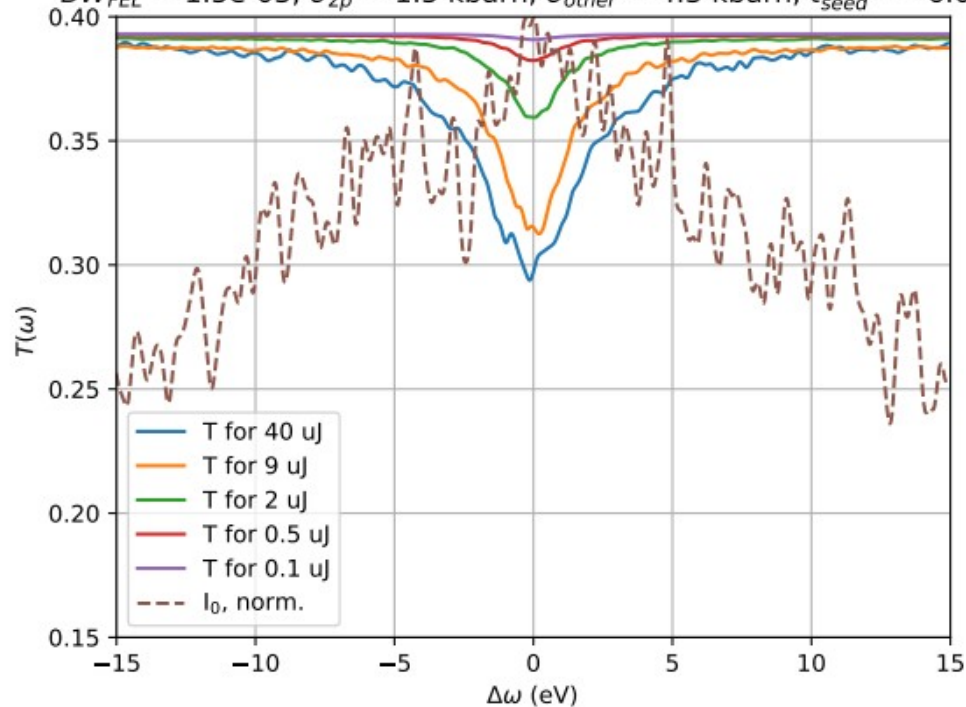


Sensitivity to pulse duration

average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 200$

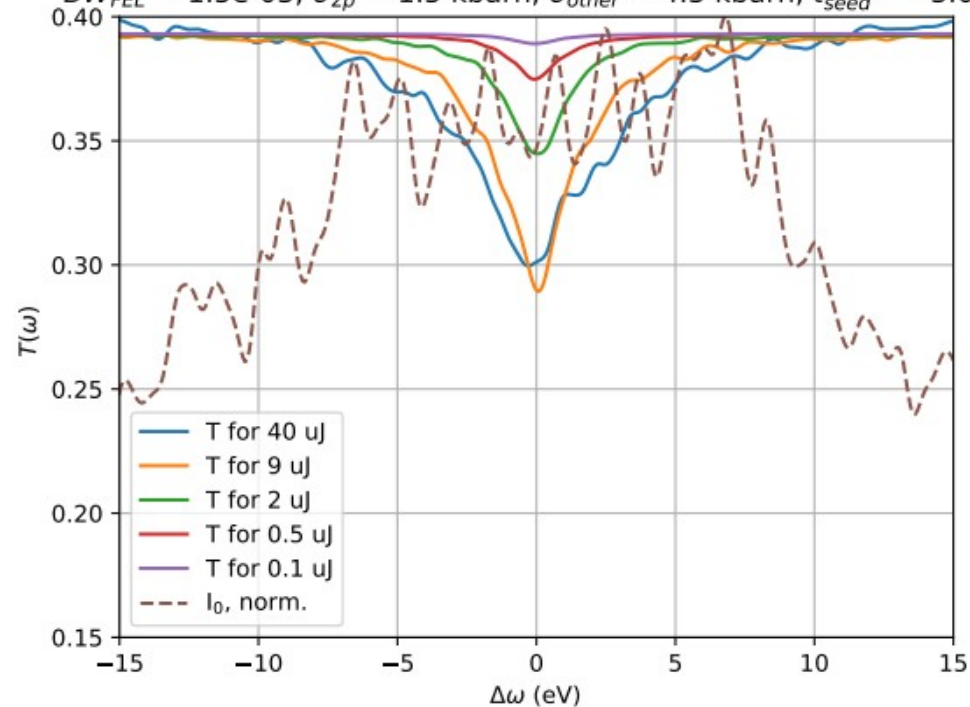
$BW_{FEL} = 1.5e-03, \sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 6.0 \text{ fs}$



average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 500$

$BW_{FEL} = 1.5e-03, \sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 3.0 \text{ fs}$

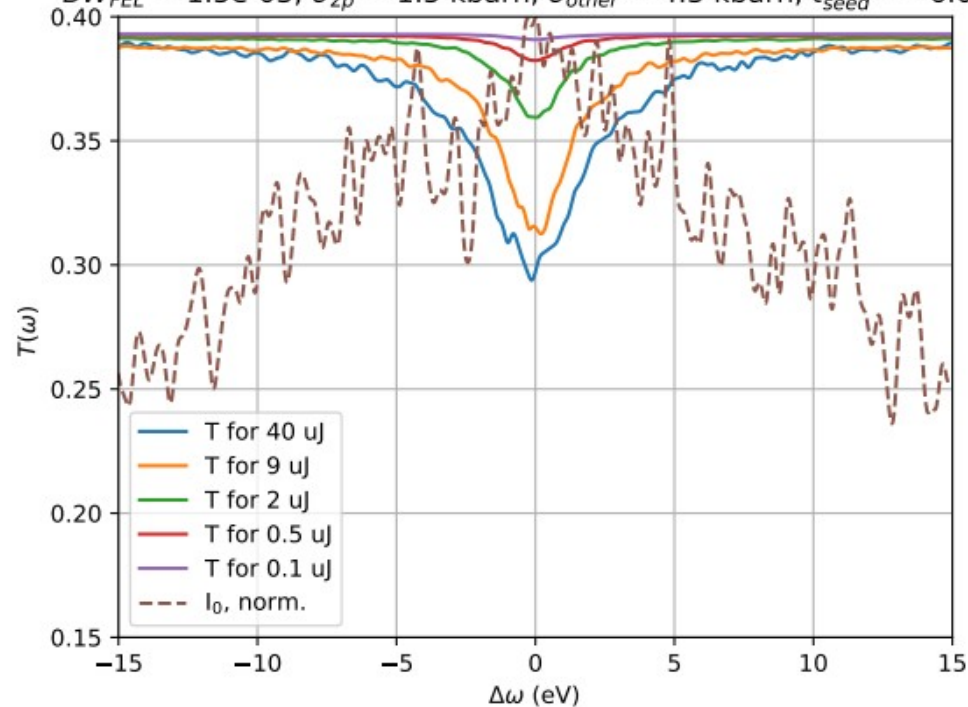


Sensitivity to pulse duration

average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 200$

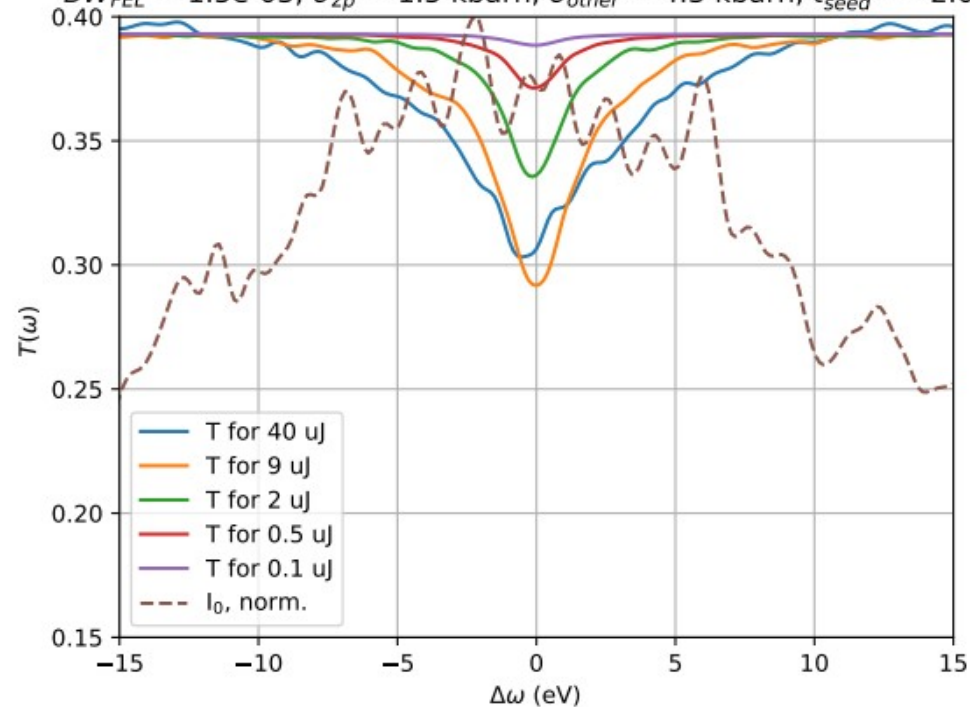
$BW_{FEL} = 1.5e-03, \sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 6.0 \text{ fs}$



average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 500$

$BW_{FEL} = 1.5e-03, \sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 2.0 \text{ fs}$

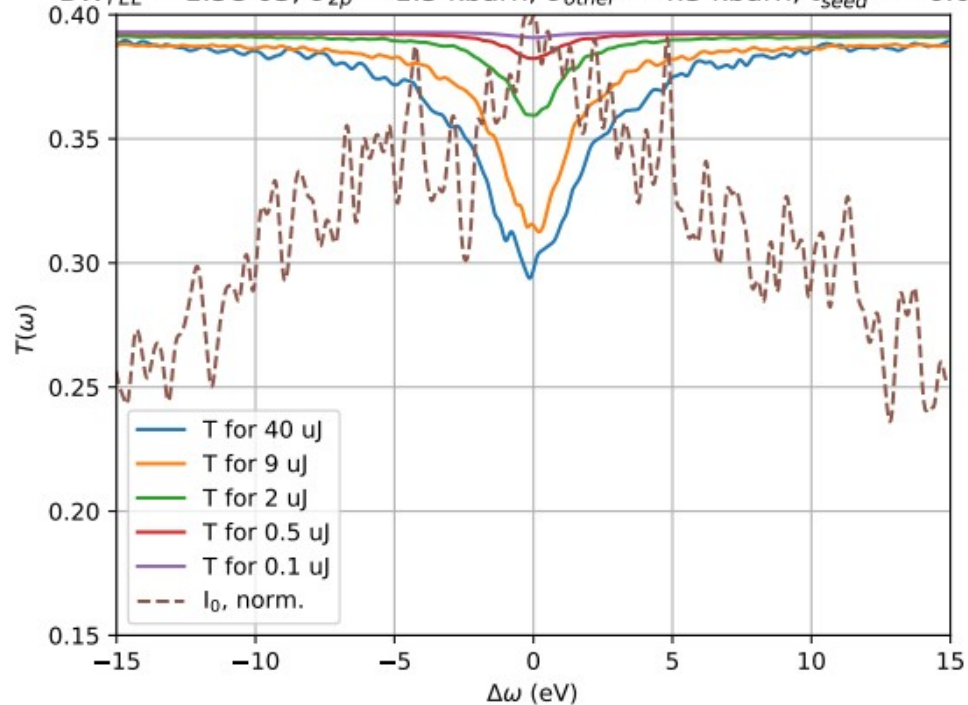


Sensitivity to pulse shape

average over 90 SASE realizations

$N_x = 8, N_y = 8, N_z = 30, N_t = 200$

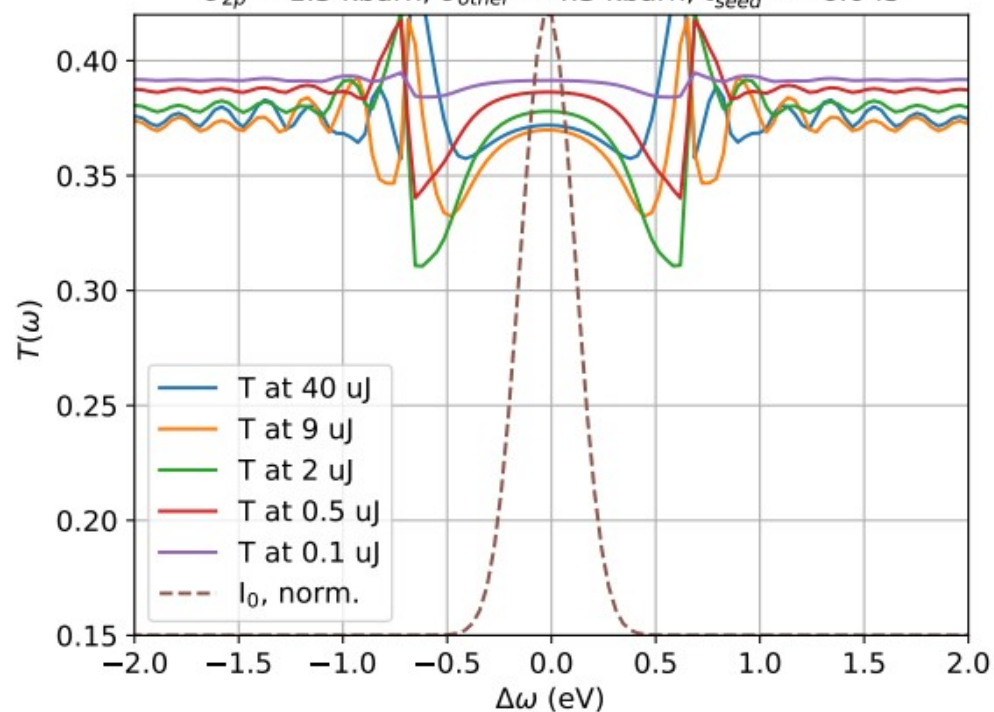
$BW_{FEL} = 1.5e-03, \sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 6.0 \text{ fs}$



average over 1 Gaussian realization

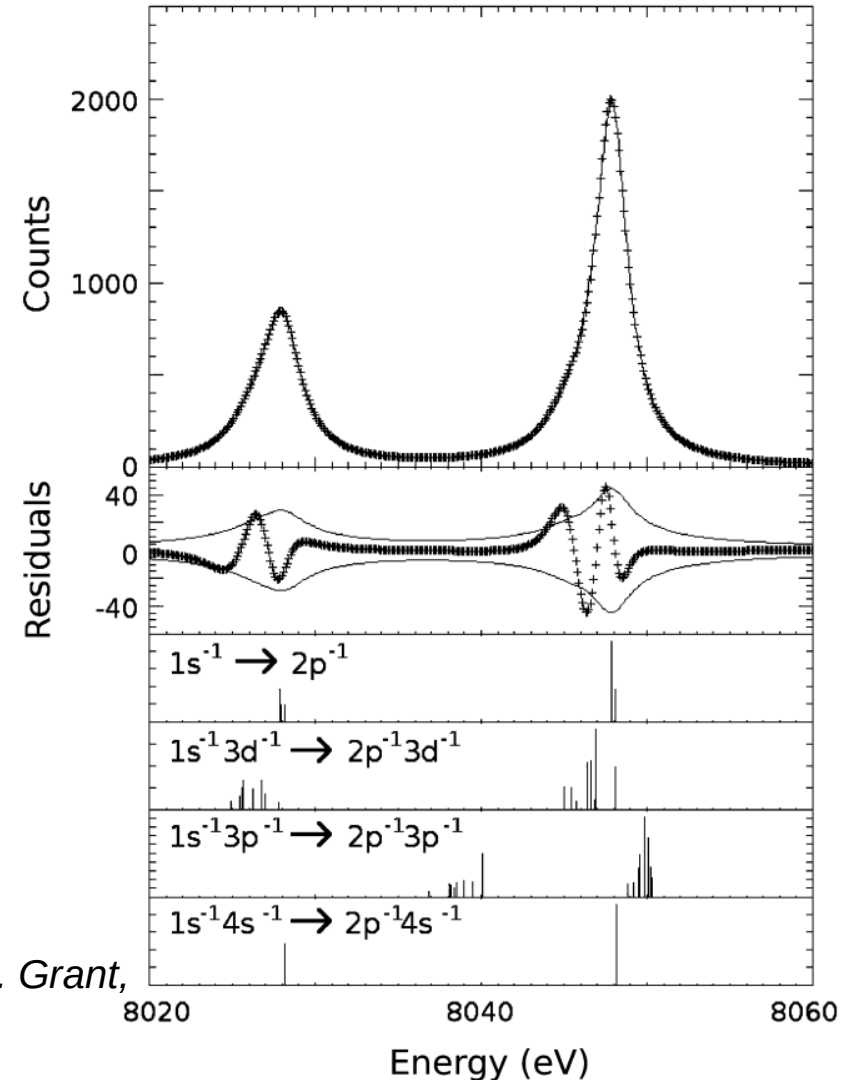
$N_x = 8, N_y = 8, N_z = 30, N_t = 400$

$\sigma_{2p} = 1.5 \text{ kbarn}, \sigma_{other} = 4.3 \text{ kbarn}, t_{seed}^{FWHM} = 6.0 \text{ fs}$



Possible sources of discrepancy

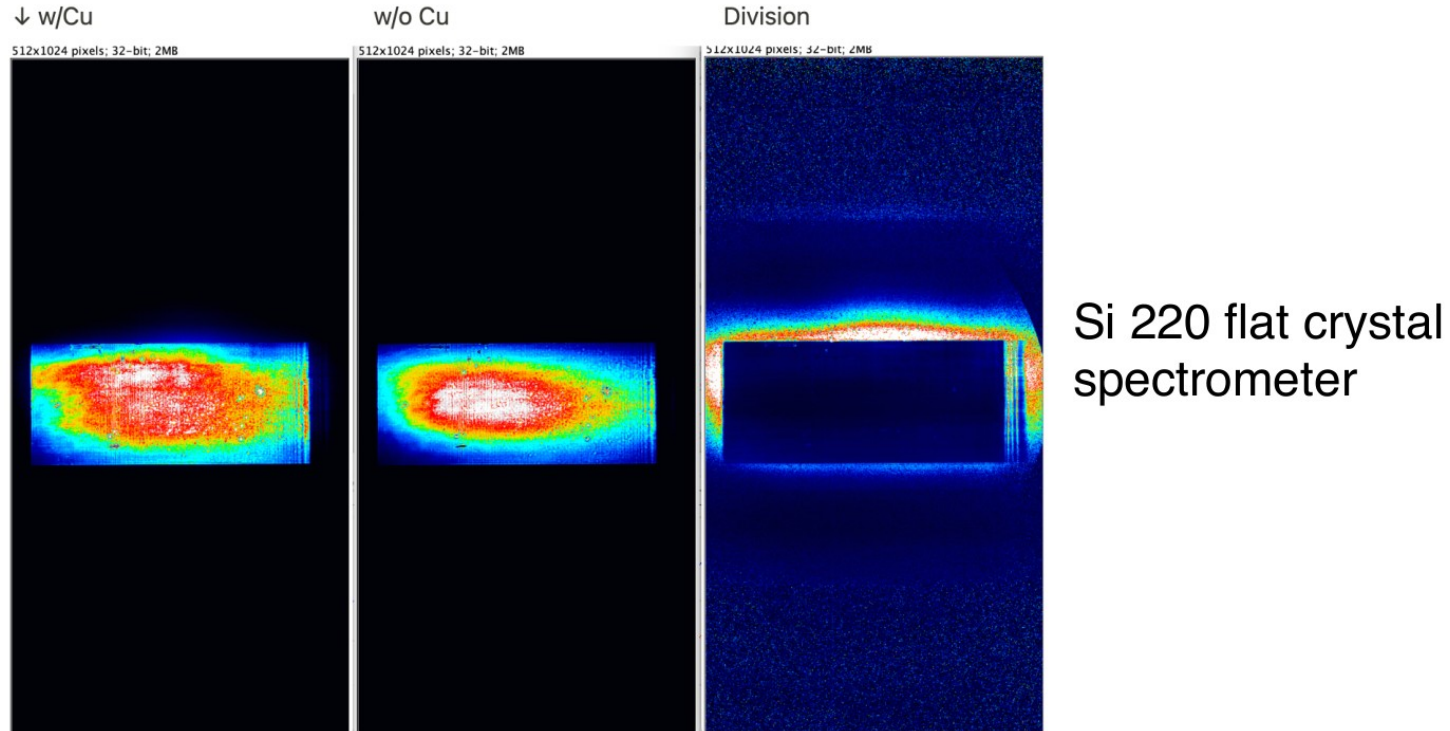
- K-alpha 2 line is not accounted for
- Electron-impact ionization – most likely, negligible
- States after $L_3 M_{45} M_{45}$ Auger decay: these states are available for further ionization and may show subsequent satellite lines in absorption. Such states may be longer-lived and thus pronounced.



*C. T. Chantler, J. A. Lowe, I. P. Grant,
10.1103/PhysRevA.82.052505*

Data from Ichiro, beamtime at July 2024

Reverse saturable absorption of copper induced by SASE pulses

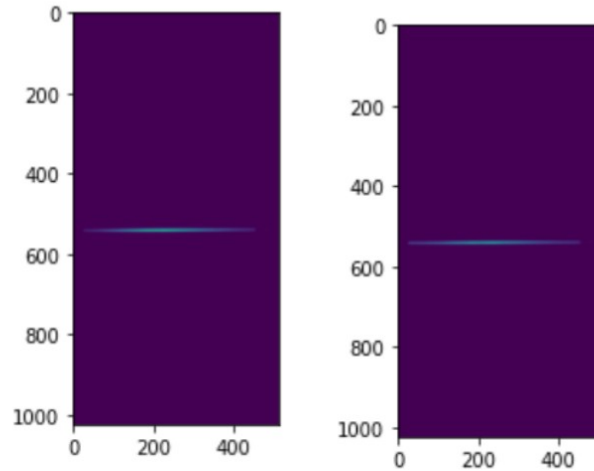


Beam divergence after transmitting copper foil seems different from original value

-> Crystal spectrometer might not be a good choice for precise evaluation of the transmittance.

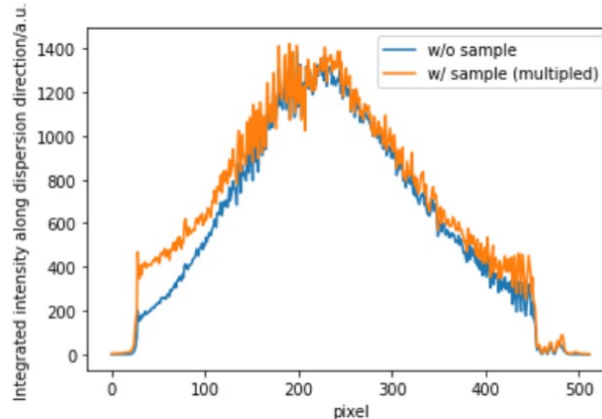
Data from Ichiro, beamtime at July 2024

RSA of copper induced by self-seeded pulses



8.048 keV=Ka1

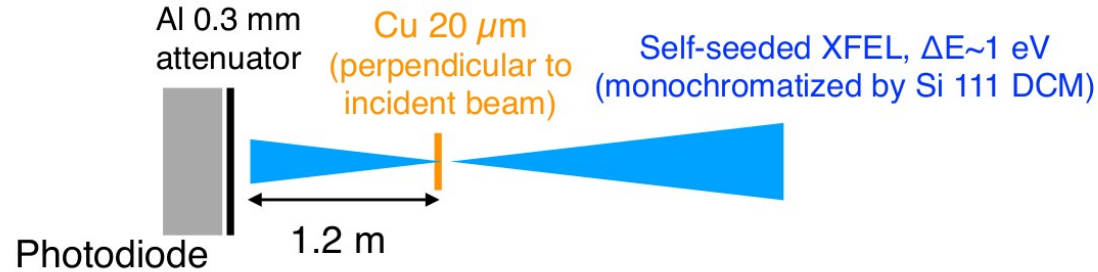
Si 220 flat crystal
spectrometer



Beam divergence becomes large
after transmitting copper foil

Data from Ichiro, beamtime at July 2024

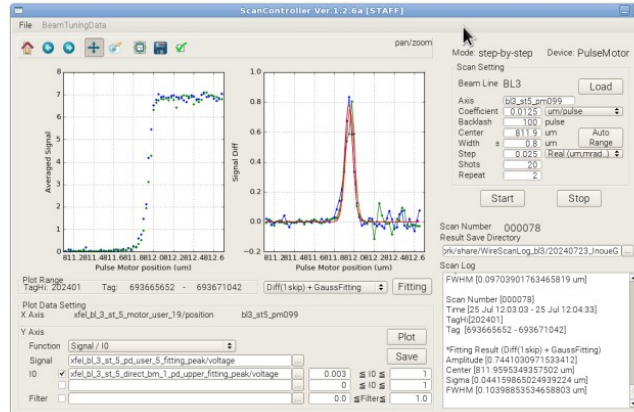
Setup



cf.

Cu foil was placed in front of photodiode when measuring transmittance of cold state.

Beam size (FWHM): 100 nm (H) x 170 nm (V)

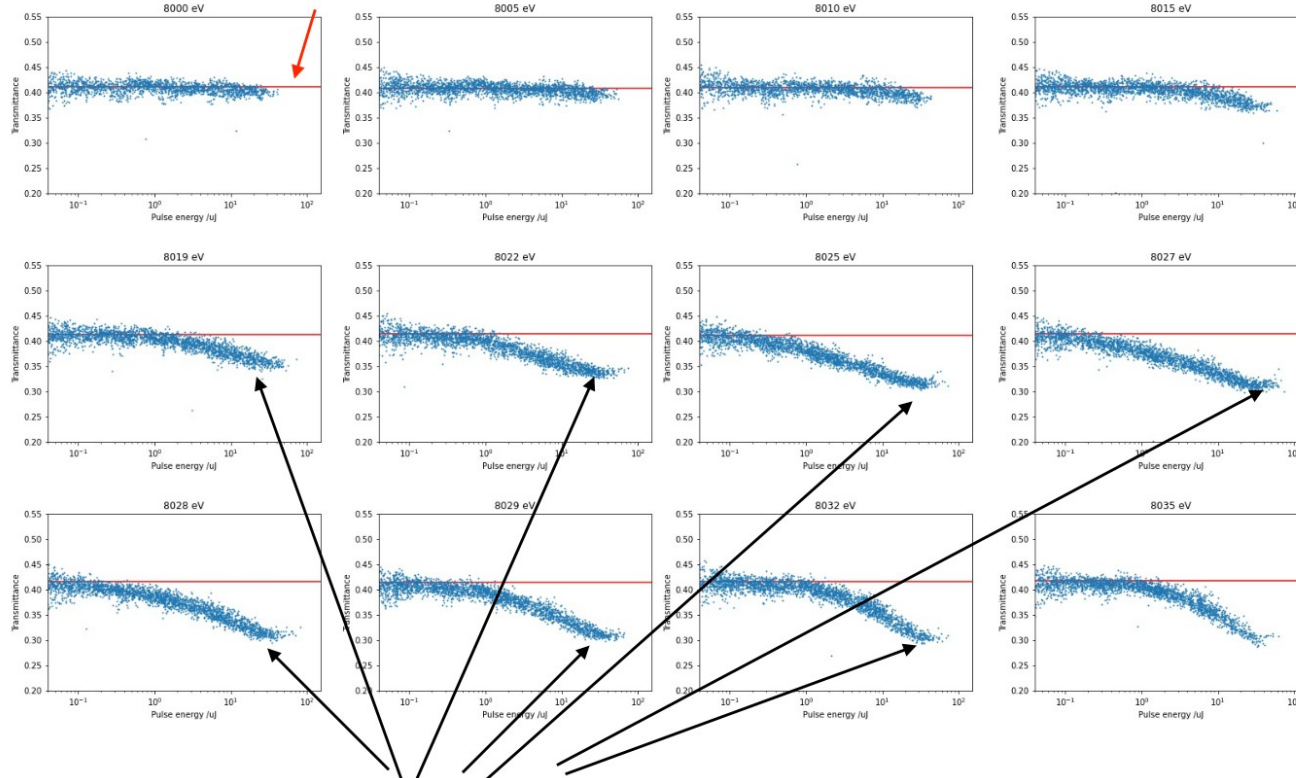


Data from Ichiro, beamtime at July 2024

Results for Cu 20 μm foil (photon energies around $\text{Ka2}=8028\text{ eV}$)

transmittance
for cold state (experimentally measured)

Each dot corresponds to single pulse

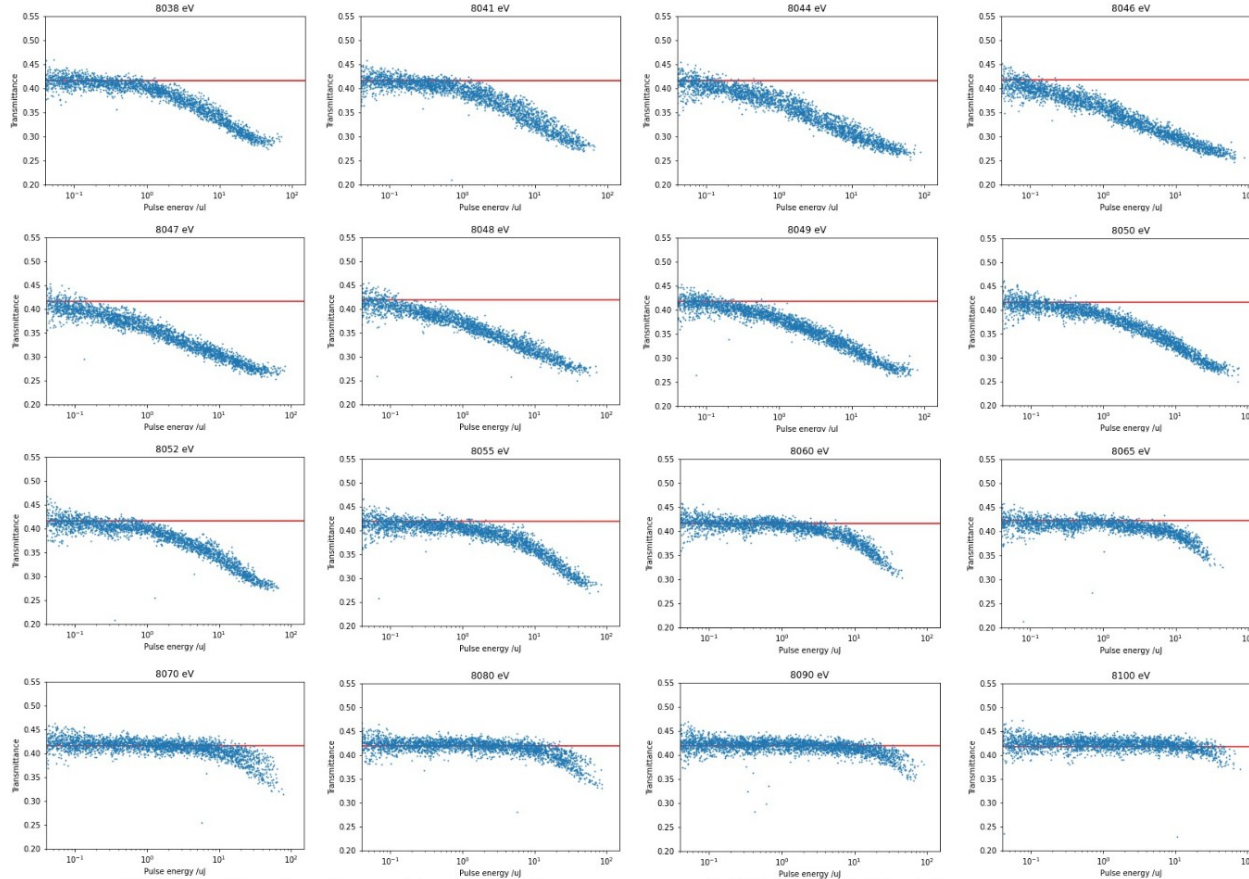


Reverse saturable absorption reached “saturation” (transmittance ~ 0.32 @ 8028 eV)

RSA was observed for photon energies higher than 8010 eV.

Data from Ichiro, beamtime at July 2024

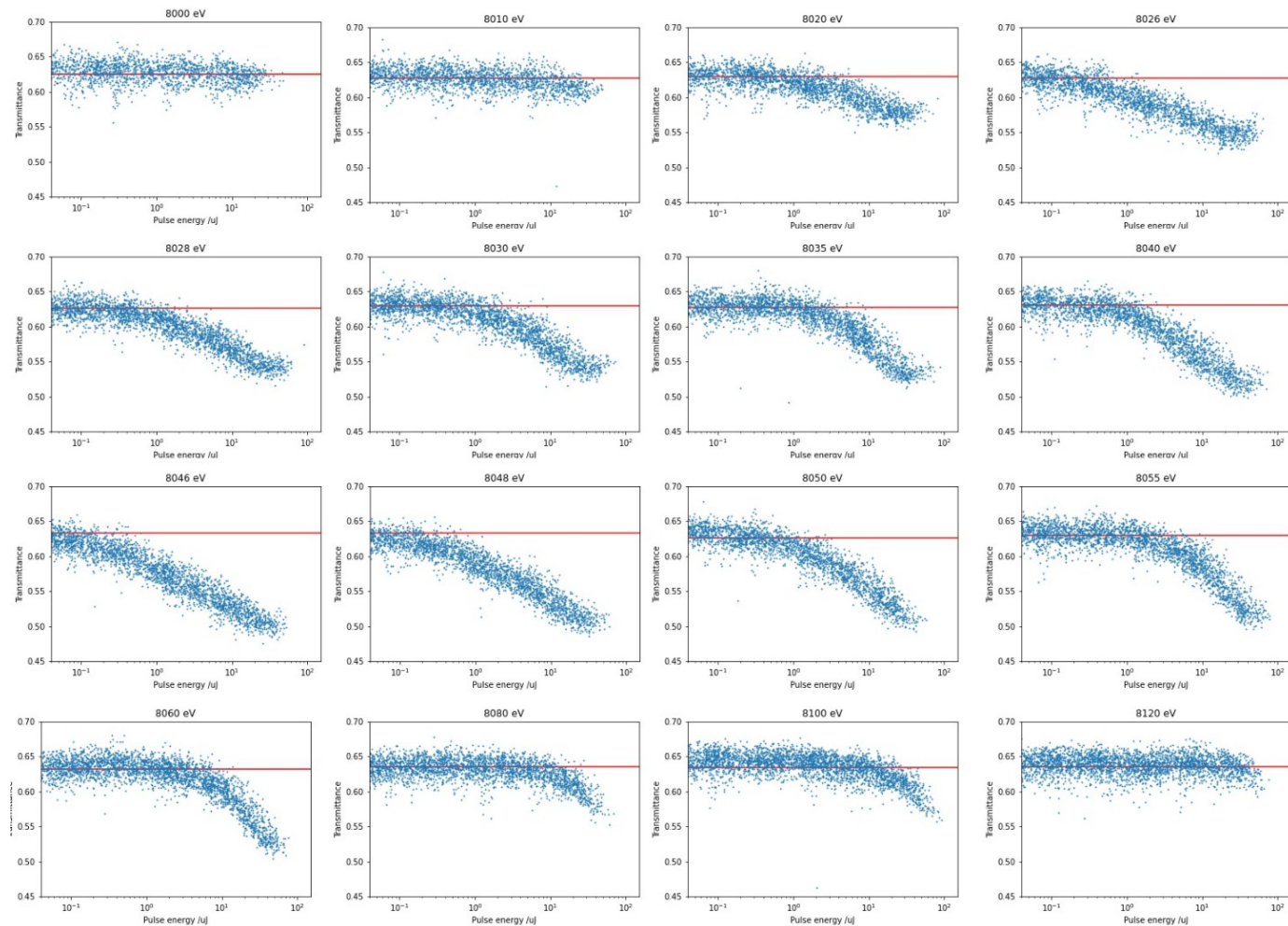
Results for Cu 20 um foil (photon energies around $K\alpha_1=8048$ eV)



“saturation” value of transmittance was ~ 0.28 @ ~ 8048 eV
RSA was observed for photon energies lower than 8090 eV.

Data from Ichiro, beamtime at July 2024

Results for Cu 10 um foil:



Data from Ichiro, beamtime at July 2024

Change in imaginary part of atomic scattering factor

Excess linear absorption coefficient due to reverse saturable absorption@8.048 keV:

$$\begin{aligned}u_{RSA} &= -\log(0.5/0.65)/10 \text{ } \mu\text{m} \\ &= 260 \text{ cm}^{-1}\end{aligned}$$

Excess absorption cross section per atom:

$$\sigma_{atom} = \frac{M}{\rho N_A} u_{RSA} = 3.2 \times 10^{-21} \text{ cm}^2$$

M: molecular mass (63.55)
ρ: density (8.96 g/cm³)
N_A: Avogadro constant (6.02 x 10²³)

According to optical theorem, change in the imaginary part of atomic scattering factor may be given by

$$f'' = \frac{-k}{4\pi r_0} \sigma_{atom} = -0.37$$

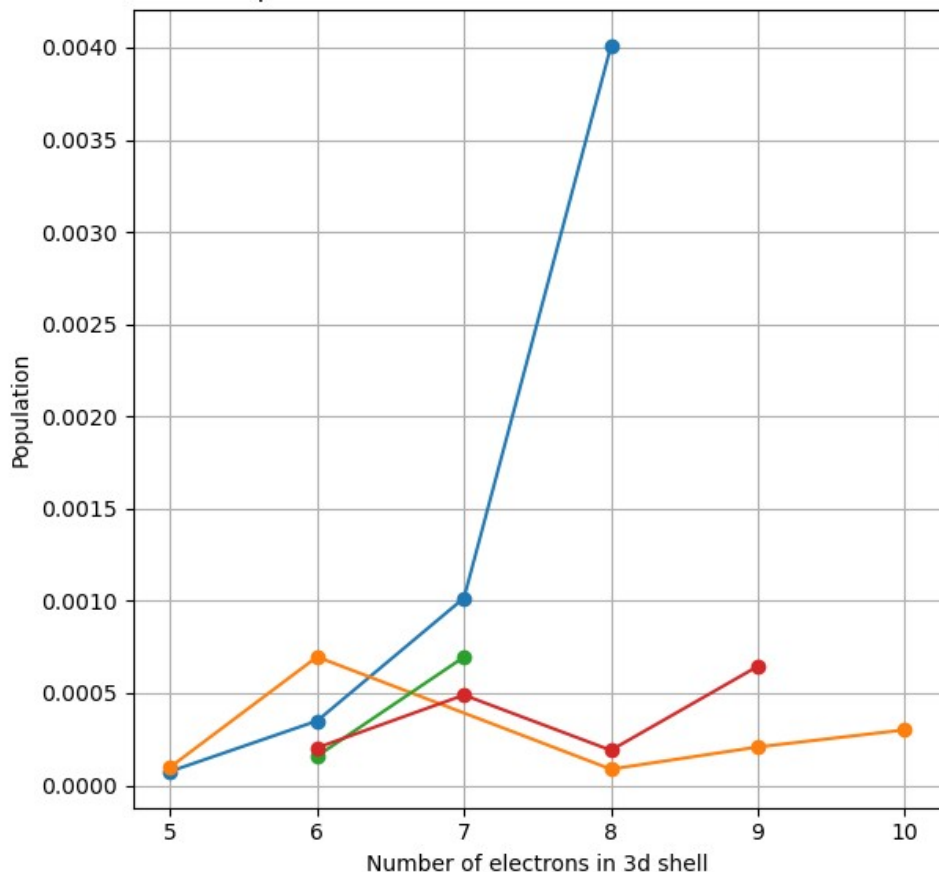
k: wave number
r₀: classical electron radius

Enhancement of atomic scattering factor by transient resonance will be very small (but might be useful for phasing)

Data from kinetic Boltzmann code

States with hole in 2p, data from Boltzmann code for 40 uJ

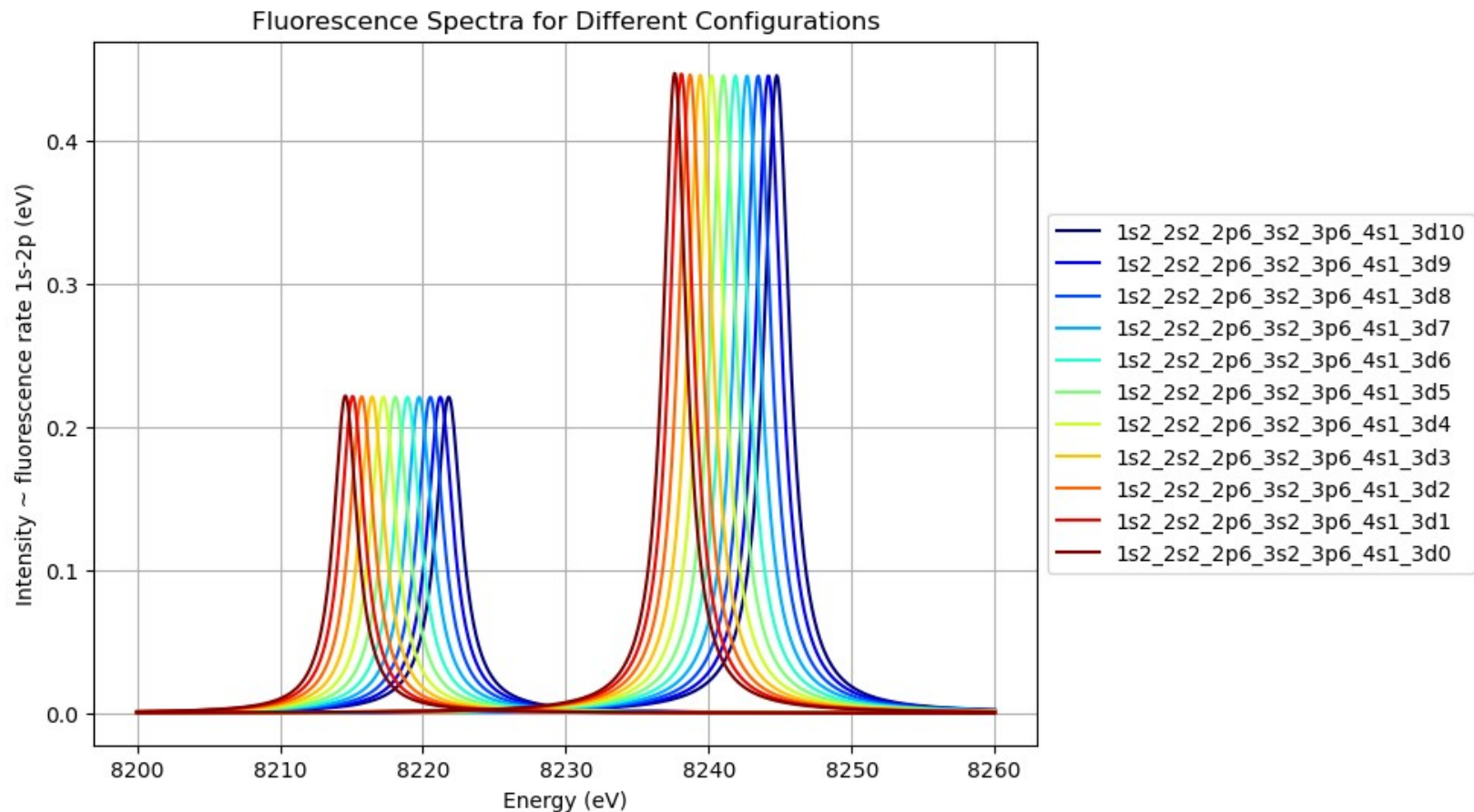
Population as a function of electrons in 3d shell



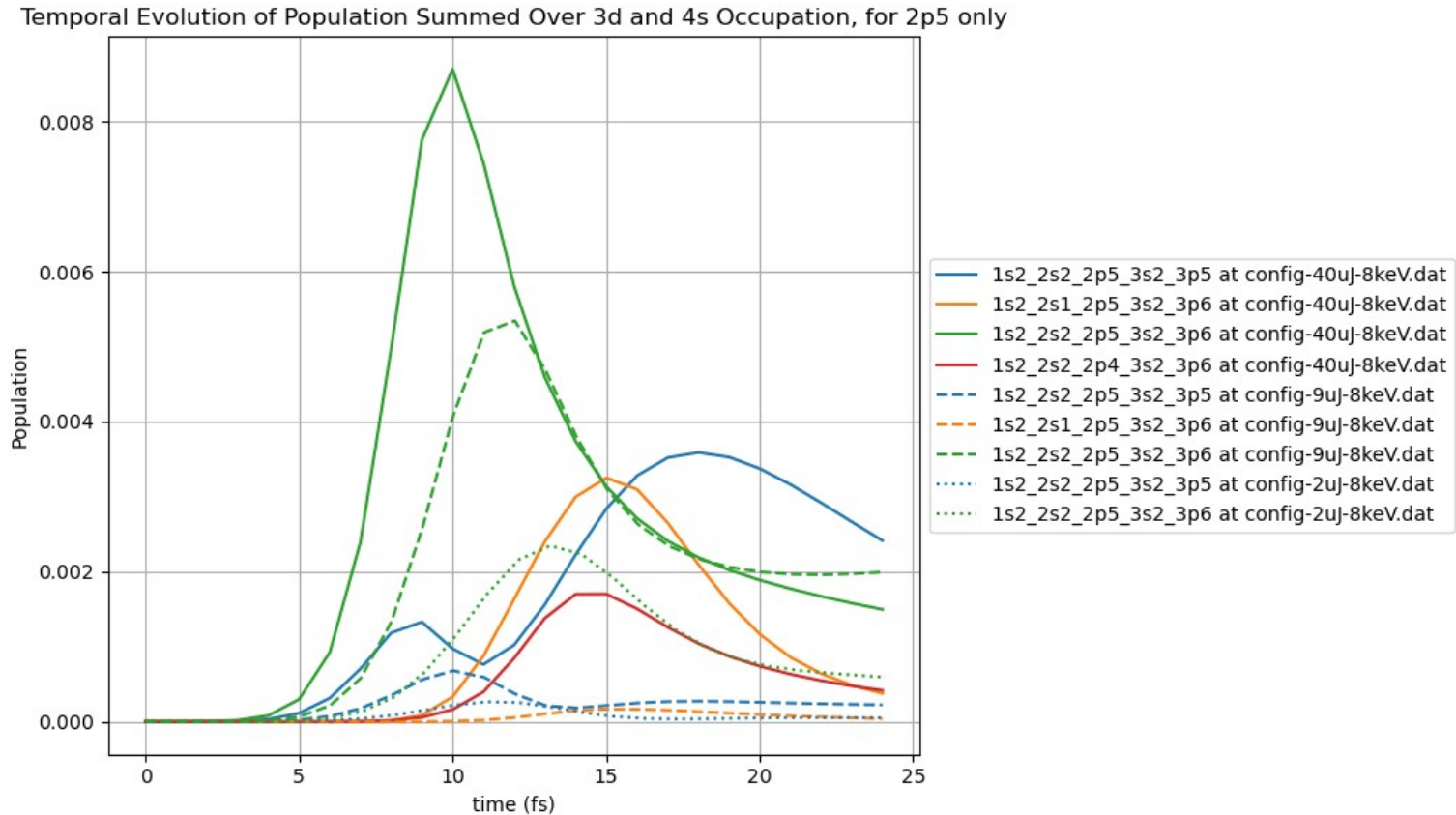
	energy_Ka1_eV	energy_Ka2_eV	rate_Ka1_eV	rate_Ka2_eV	width_1s_eV	width_2p_eV
1s2_2s2_2p6_3s2_3p6_4s1_3d8	8243.46	8220.52	0.445189	0.220741	1.543735	0.659600
1s2_2s2_2p6_3s2_3p6_4s1_3d7	8242.68	8219.74	0.445135	0.220714	1.551620	0.655561
1s2_2s2_2p6_3s2_3p5_4s1_3d6	8244.88	8221.89	0.446025	0.221152	1.533879	0.628813
1s2_2s2_2p5_3s2_3p6_4s1_3d7	8277.62	8253.97	0.343661	0.227148	1.946224	1.437282
1s2_2s1_2p6_3s2_3p6_4s1_3d9	8268.63	8245.17	0.454702	0.225422	5.663179	4.895058
1s2_2s1_2p6_3s2_3p6_4s1_3d7	8267.06	8243.60	0.454667	0.225404	2.118703	3.812288
1s2_2s2_2p6_3s2_3p6_4s1_3d6	8241.86	8218.92	0.445132	0.220712	1.560780	0.641060
1s2_2s2_2p6_3s2_3p5_4s1_3d10	8247.55	8224.56	0.446052	0.221166	4.648869	3.499216
1s2_2s2_2p6_3s2_3p5_4s1_3d9	8246.95	8223.96	0.445978	0.221130	3.649866	0.844403
1s2_2s1_2p6_3s2_3p6_4s1_3d6	8266.26	8242.80	0.454749	0.225444	2.126039	1.374364
1s2_2s1_2p6_3s2_3p6_4s1_3d8	8267.86	8244.40	0.454659	0.225399	5.326939	4.618965
1s2_2s2_2p5_3s2_3p6_4s1_3d6	8276.81	8253.16	0.343721	0.227188	1.901719	1.352097
1s2_2s2_2p6_3s2_3p5_4s1_3d5	8244.19	8221.20	0.446180	0.221229	1.543484	0.591389
1s2_2s2_2p6_3s2_3p5_4s1_3d8	8246.28	8223.30	0.445937	0.221110	1.684728	0.842682
1s2_2s2_2p6_3s2_3p6_4s1_3d5	8241.02	8218.08	0.445197	0.220744	1.571107	0.614073

- 1s2_2s2_2p5_3s2_3p6_4s1_3d0
- 1s2_2s2_2p5_3s2_3p5_4s1_3d0
- 1s2_2s2_2p4_3s2_3p6_4s1_3d0
- 1s2_2s1_2p5_3s2_3p6_4s1_3d0

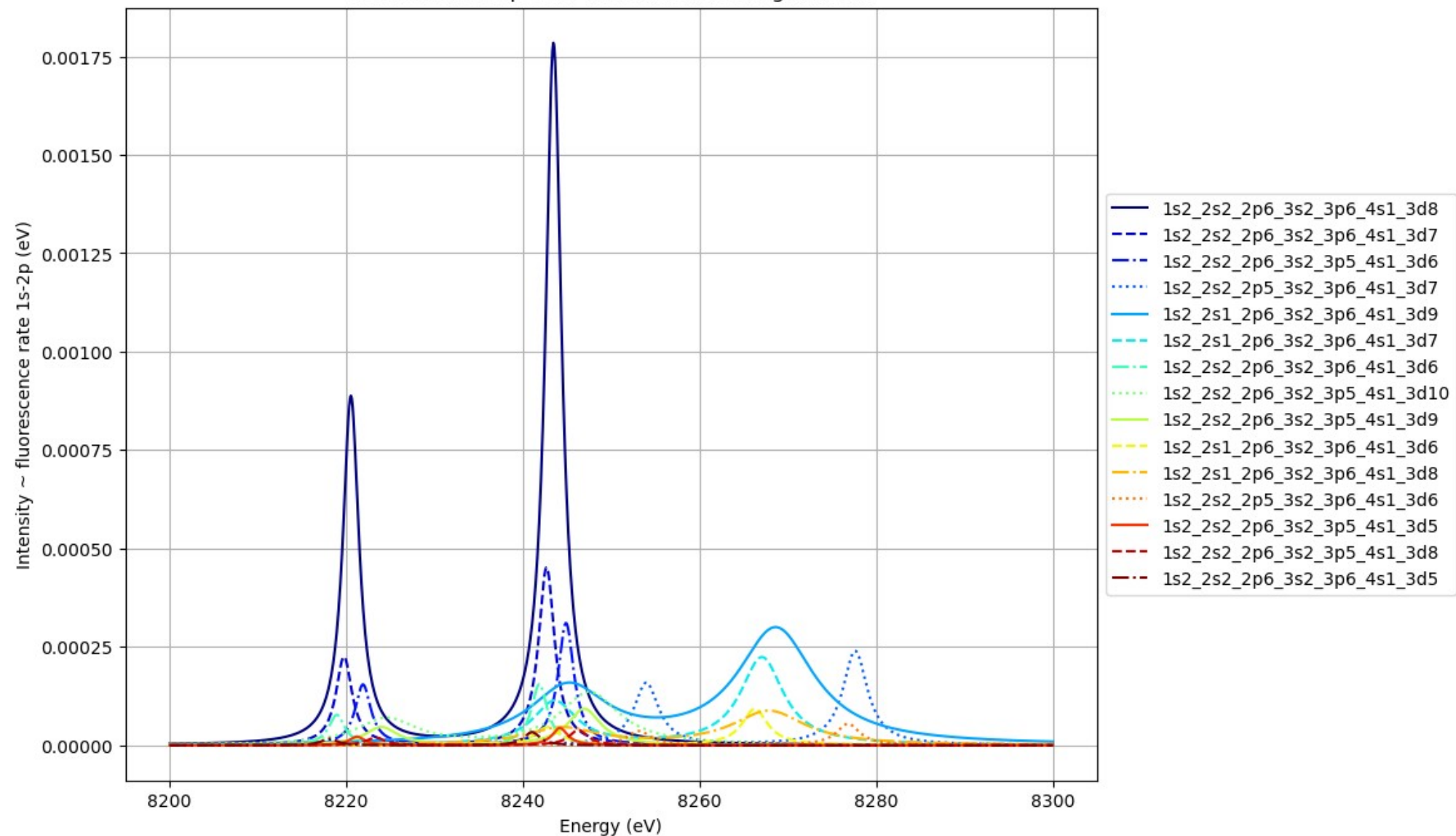
XATOM positions of fluorescence lines for 3d states



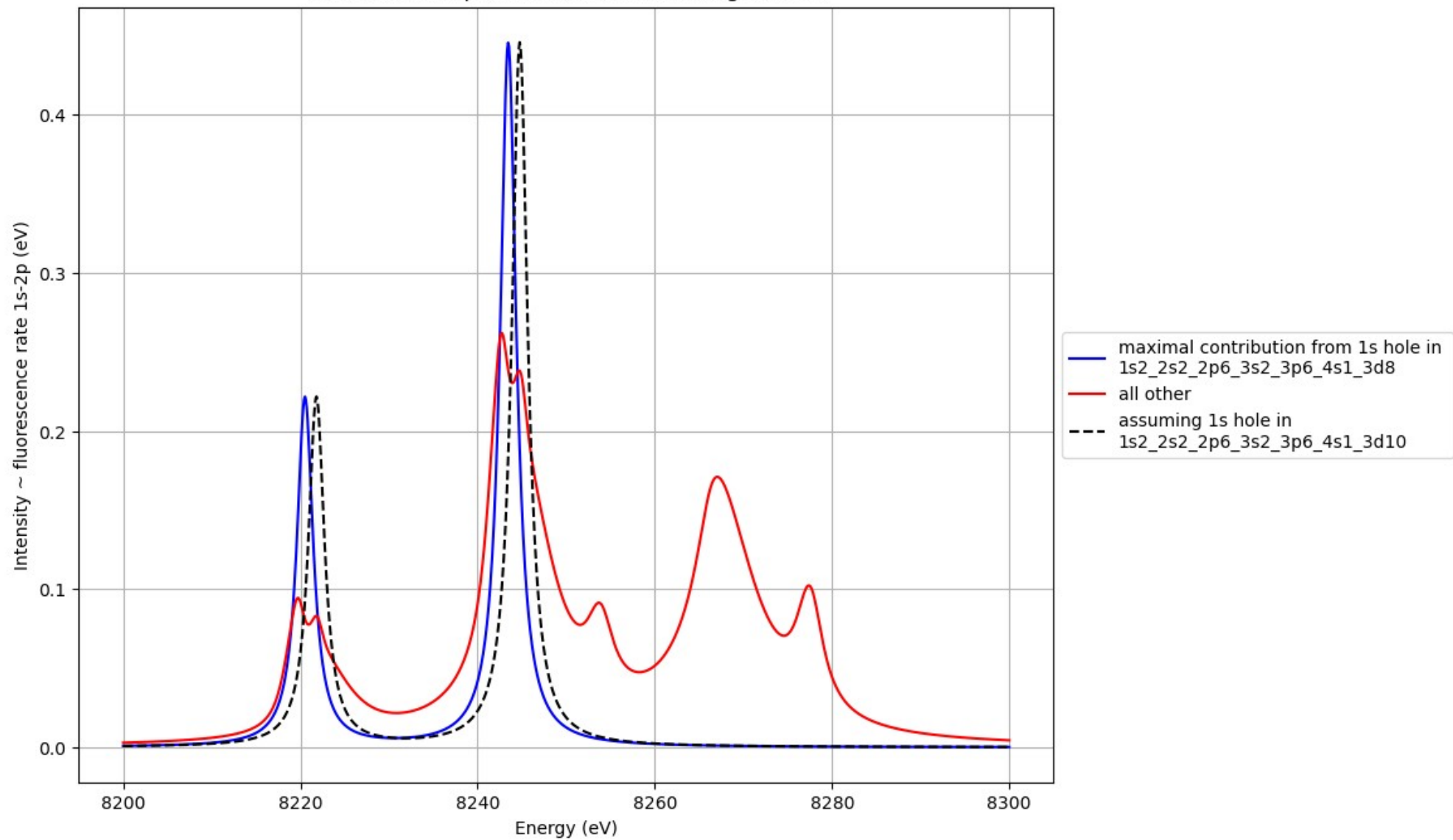
Role of other states (hole in 2s1, 2p4, 3p5), data from Boltzmann code

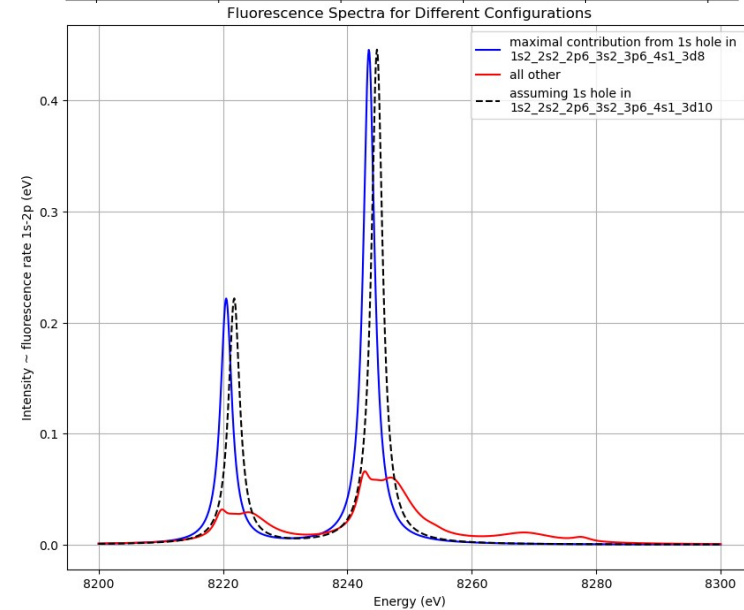
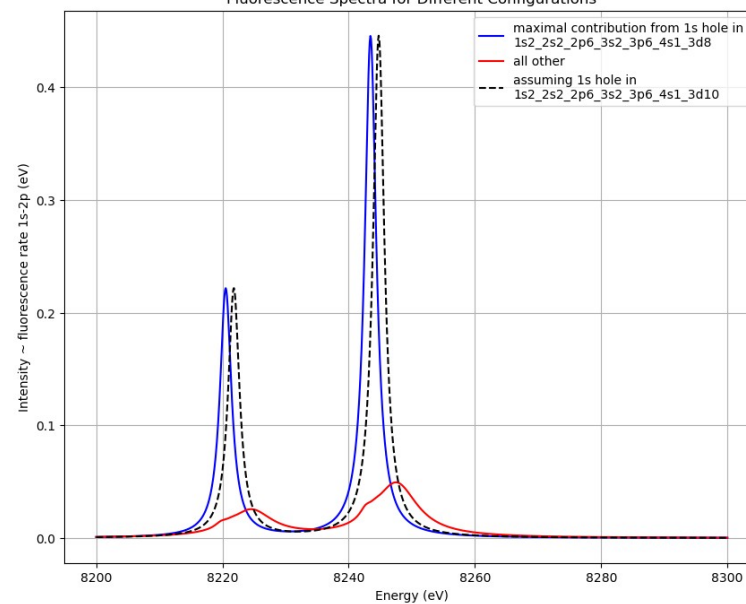
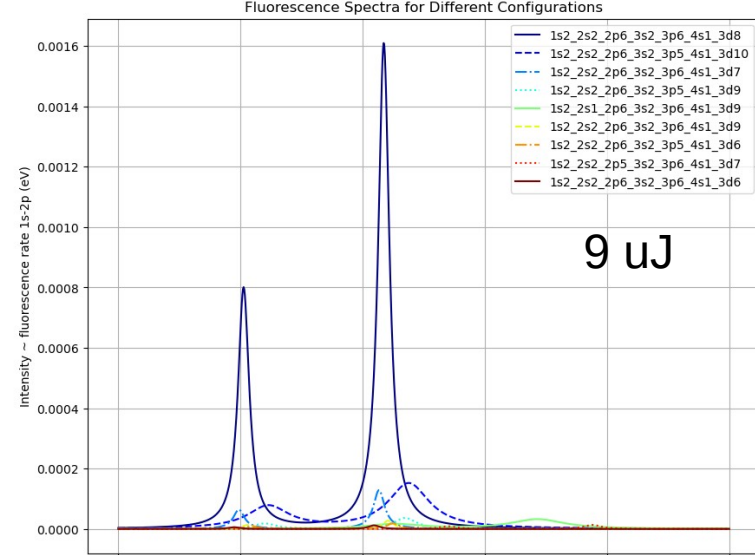
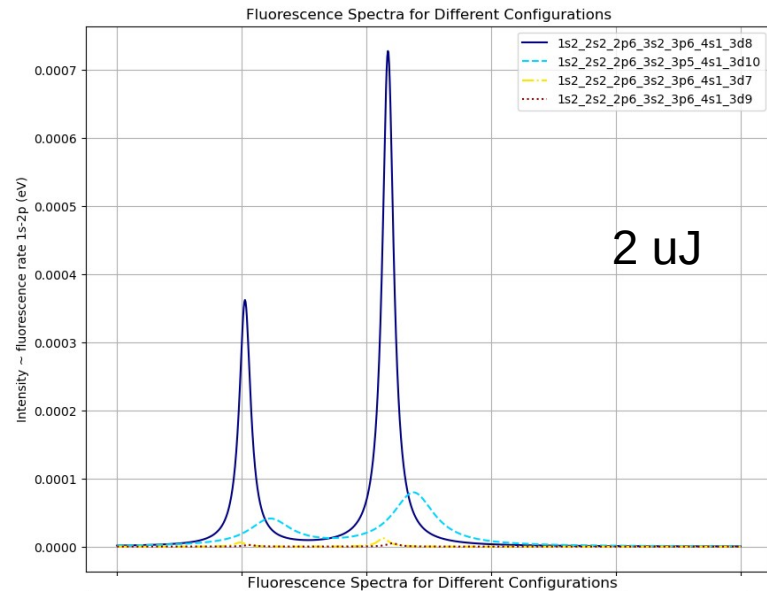


Fluorescence Spectra for Different Configurations

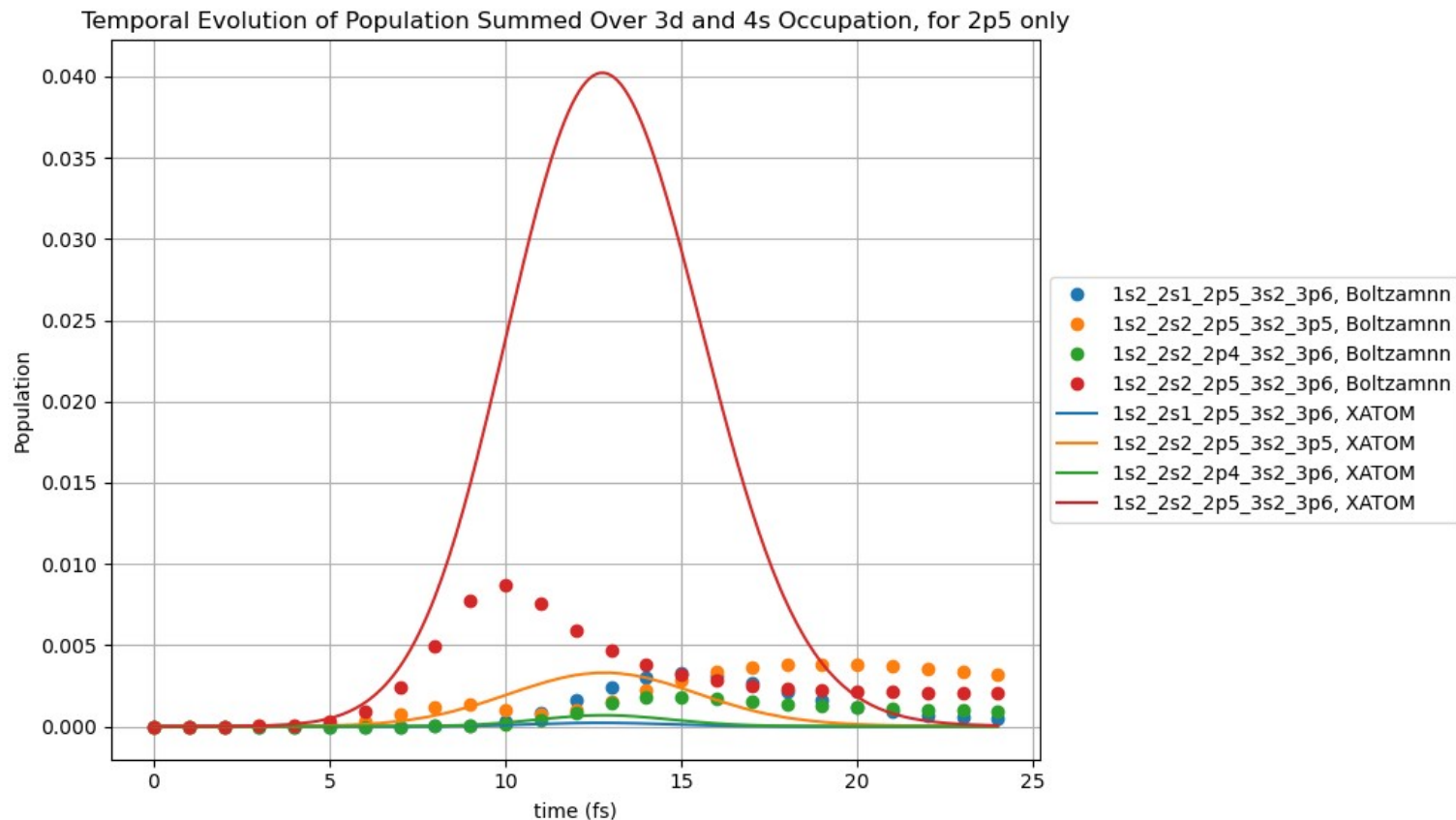


Fluorescence Spectra for Different Configurations



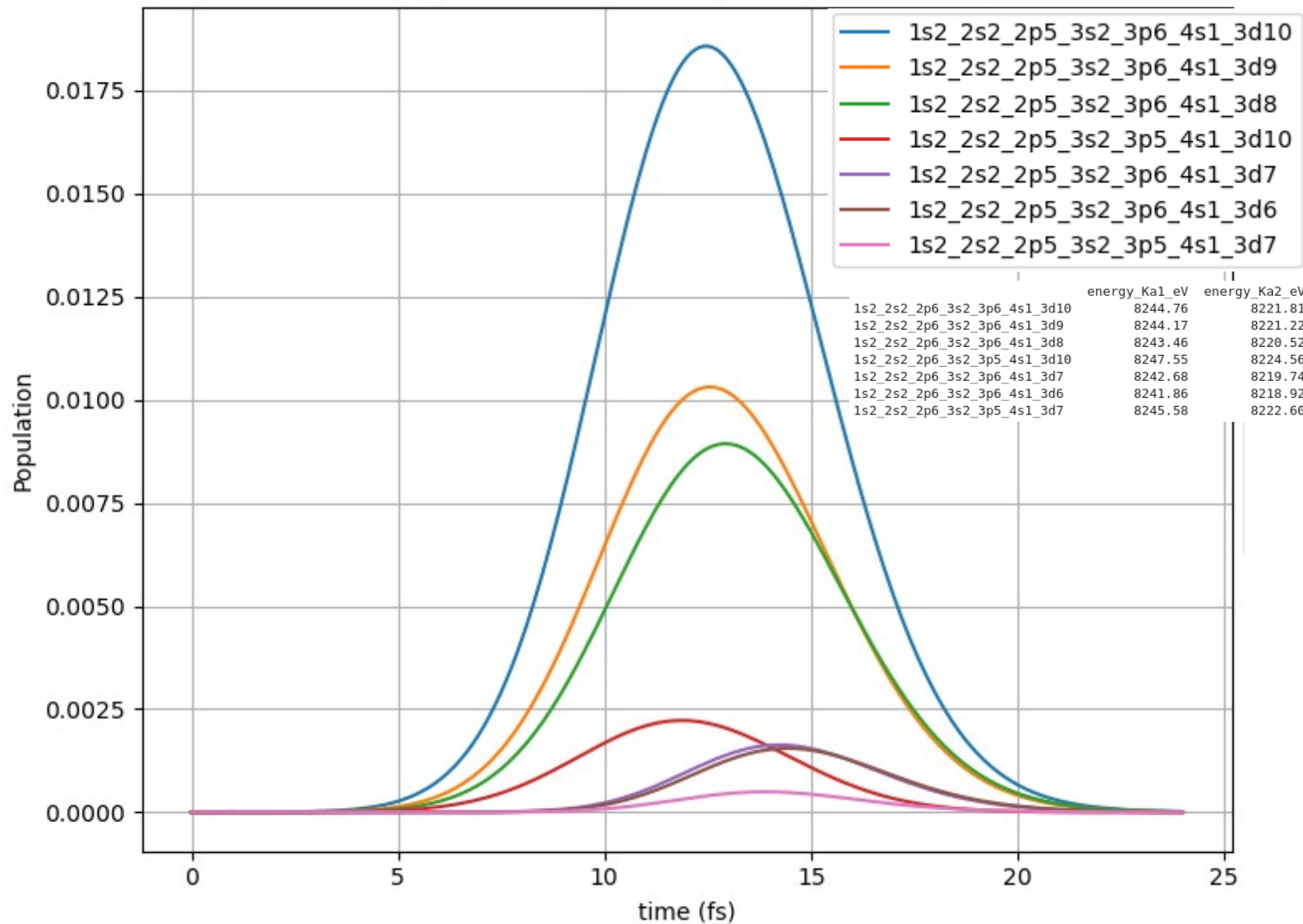


Boltzmann and XATOM code are quite different for the states with 2p holes



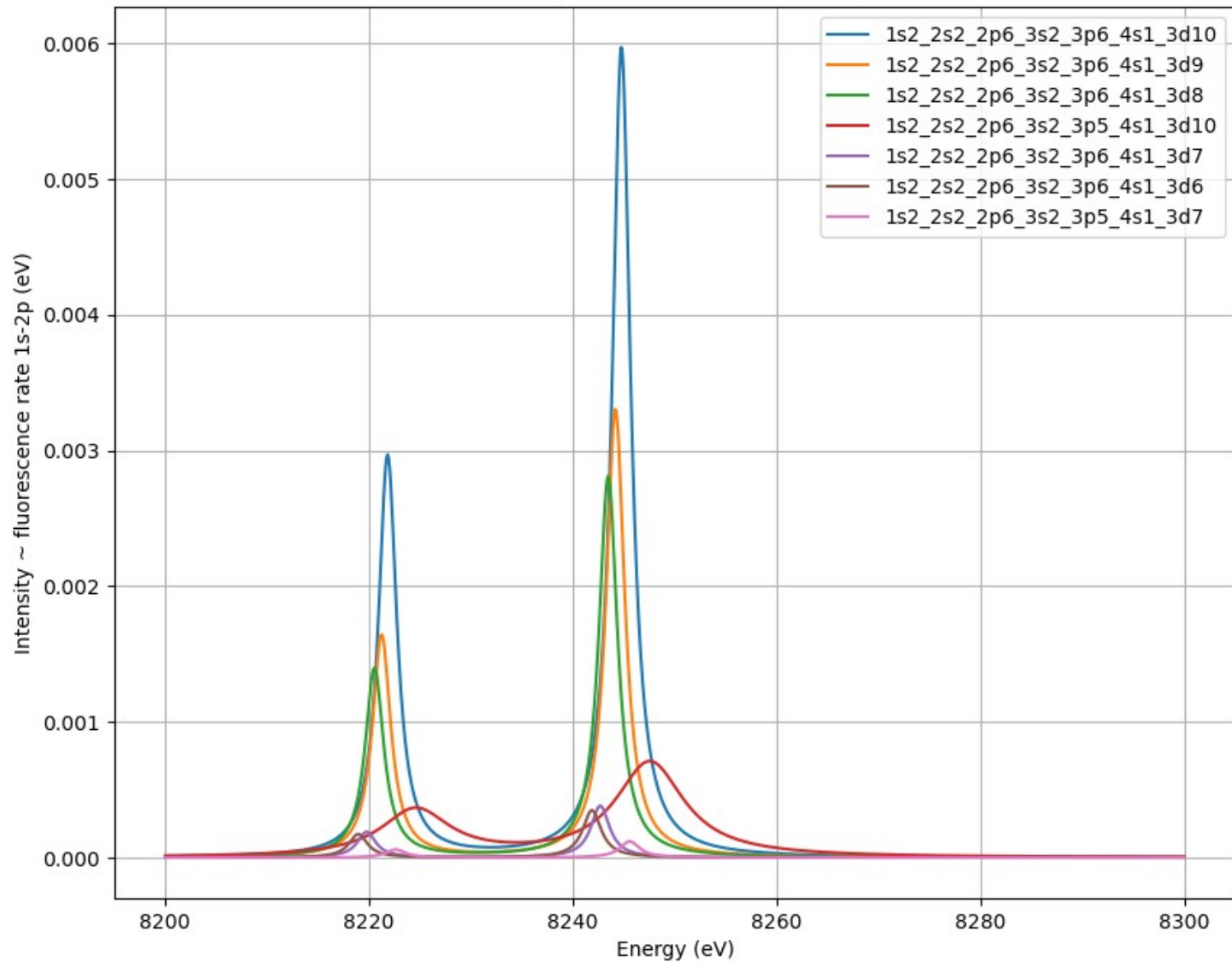
Non-relativistic (no spin-orbit) calculations with XATOM

Temporal Evolution of Population for 2p5 only



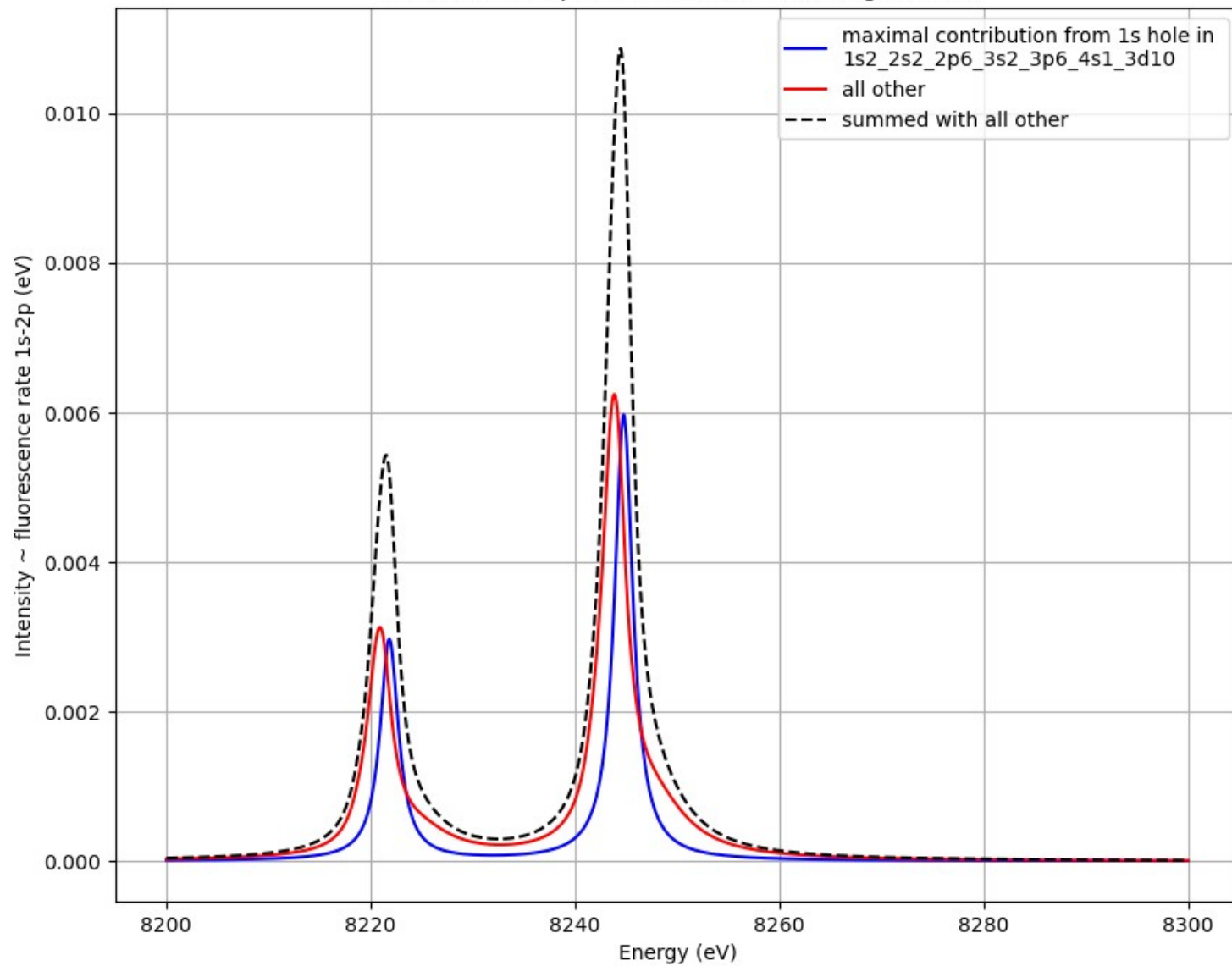
	energy_Ka1_eV	energy_Ka2_eV	rate_Ka1_eV	rate_Ka2_eV	width_1s_eV	width_2p_eV
1s2_2s2_2p6_3s2_3p6_4s1_3d10	8244.76	8221.81	0.445382	0.220837	1.532243	0.650702
1s2_2s2_2p6_3s2_3p6_4s1_3d9	8244.17	8221.22	0.445279	0.220785	1.537231	0.655461
1s2_2s2_2p6_3s2_3p6_4s1_3d8	8243.46	8220.52	0.445189	0.220741	1.543735	0.659600
1s2_2s2_2p6_3s2_3p5_4s1_3d10	8247.55	8224.56	0.446052	0.221166	4.648869	3.499216
1s2_2s2_2p6_3s2_3p6_4s1_3d7	8242.68	8219.74	0.445135	0.220714	1.551620	0.655561
1s2_2s2_2p6_3s2_3p6_4s1_3d6	8241.86	8218.92	0.445132	0.220712	1.560780	0.641060
1s2_2s2_2p6_3s2_3p5_4s1_3d7	8245.58	8222.60	0.445948	0.221115	1.681247	0.652588

Fluorescence Spectra for Different Configurations

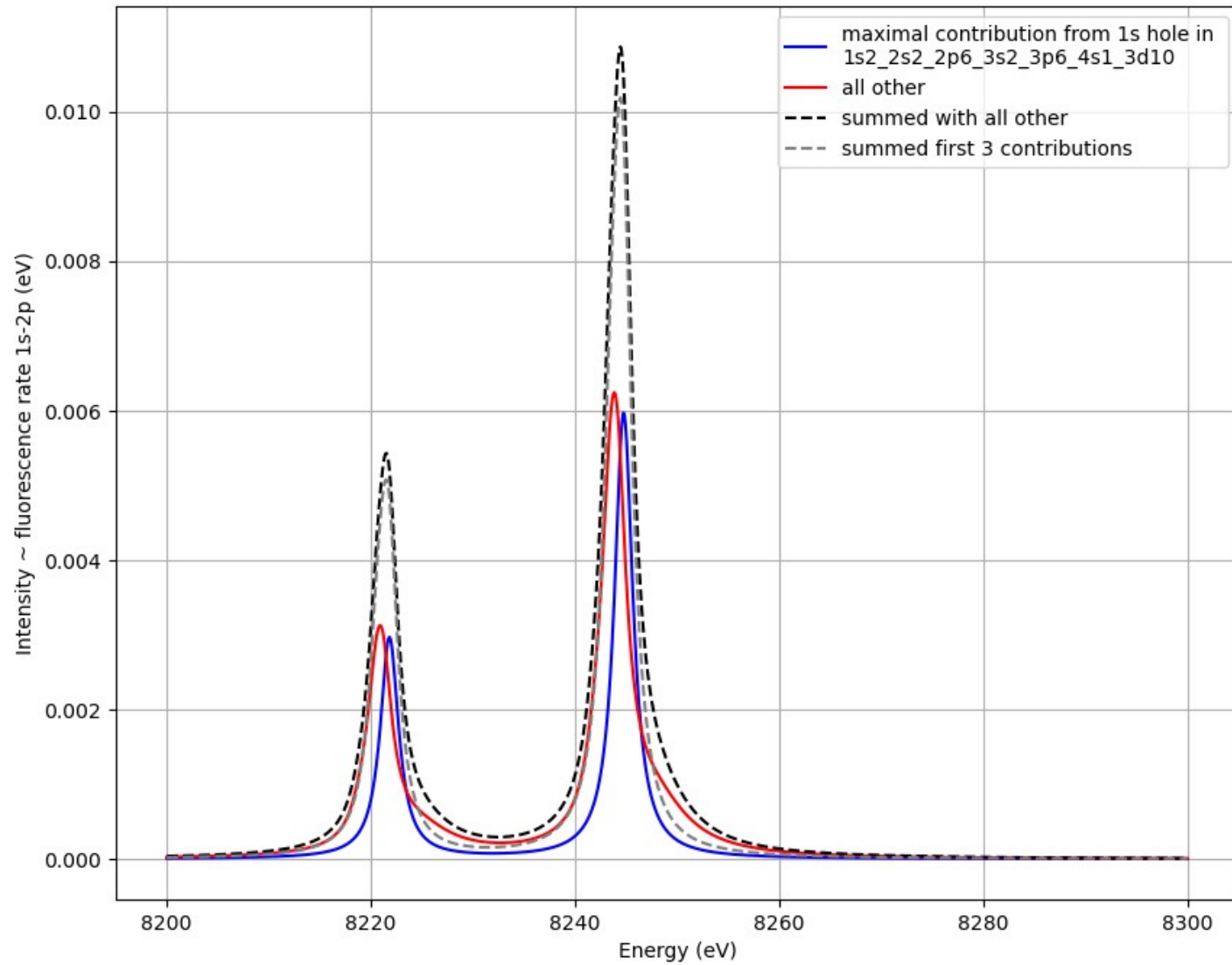


with hole in 1s

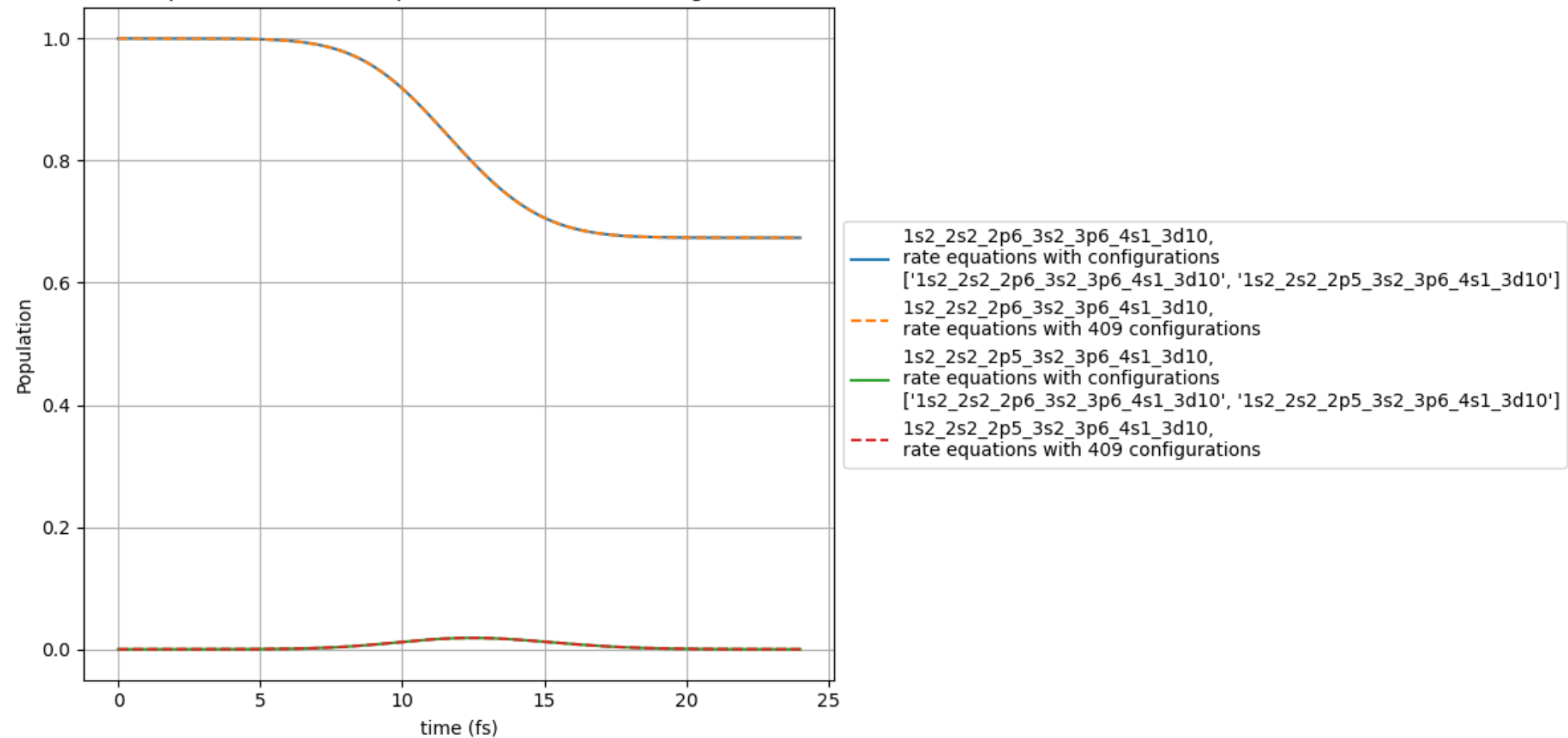
Fluorescence Spectra for Different Configurations



Fluorescence Spectra for Different Configurations



Temporal Evolution of Population for Selected Configurations



Pathways to 1s2_2s2_2p5_3s2_3p6_4s1_3d10:

1s2_2s2_2p6_3s2_3p6_4s1_3d10
| P, 39.77 fs, br=0.41
1s2_2s2_2p5_3s2_3p6_4s1_3d10
pathway branching ratio = 0.4068

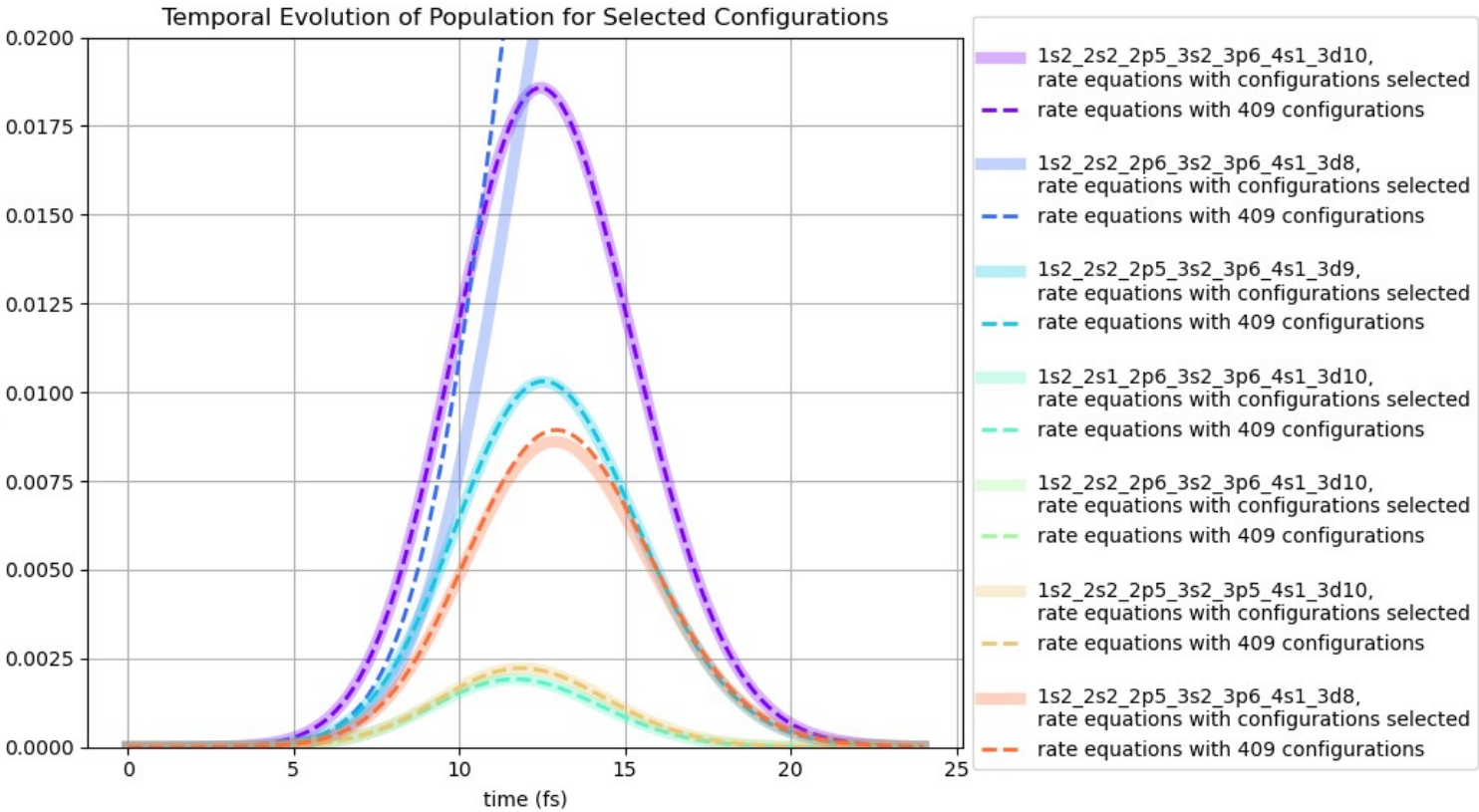
Pathways to 1s2_2s2_2p5_3s2_3p6_4s1_3d9:

1s2_2s2_2p6_3s2_3p6_4s1_3d10
| P, 33.55 fs, br=0.48
1s2_2s1_2p6_3s2_3p6_4s1_3d10
| A, 0.17 fs, br=0.46
1s2_2s2_2p5_3s2_3p6_4s1_3d9
pathway branching ratio = 0.2225

Pathways to 1s2_2s2_2p5_3s2_3p6_4s1_3d8:

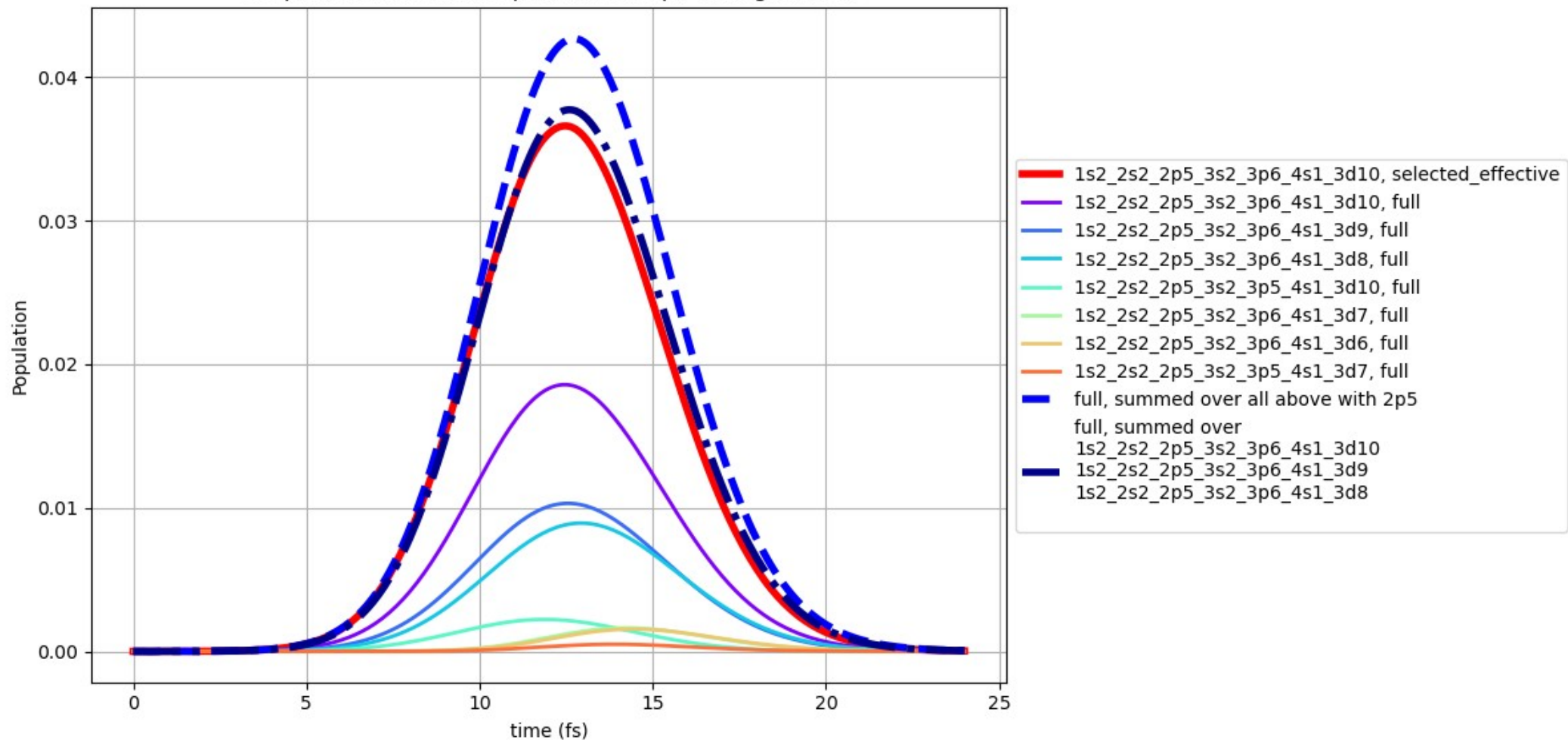
1s2_2s2_2p6_3s2_3p6_4s1_3d10
| P, 33.55 fs, br=0.48
1s2_2s1_2p6_3s2_3p6_4s1_3d10
| A, 0.16 fs, br=0.47
1s2_2s2_2p5_3s2_3p5_4s1_3d10
| A, 0.25 fs, br=0.76
1s2_2s2_2p5_3s2_3p6_4s1_3d8
pathway branching ratio = 0.1725

1s2_2s2_2p6_3s2_3p6_4s1_3d10
| P, 39.77 fs, br=0.41
1s2_2s2_2p5_3s2_3p6_4s1_3d10
| A, 2.74 fs, br=0.34
1s2_2s2_2p6_3s2_3p6_4s1_3d8
| P, 39.38 fs, br=0.40
1s2_2s2_2p5_3s2_3p6_4s1_3d8
pathway branching ratio = 0.0558

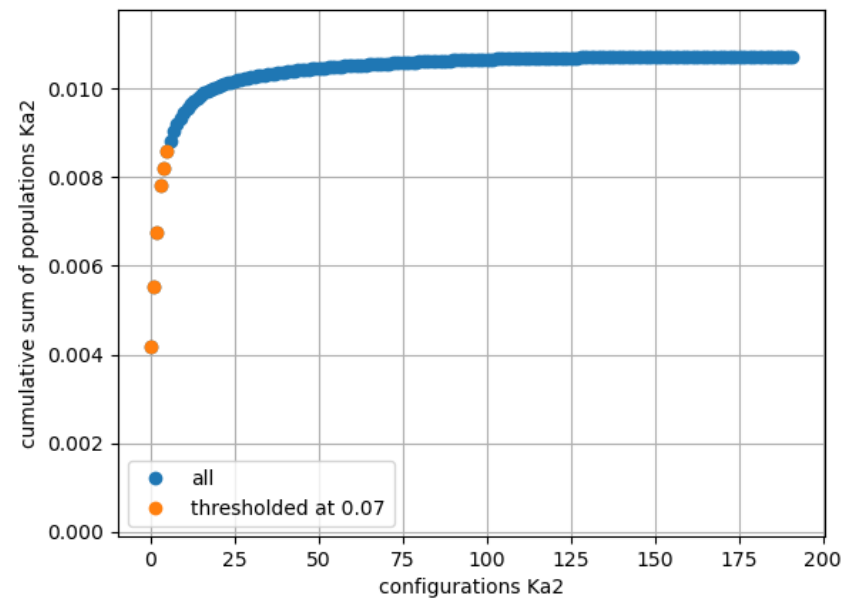
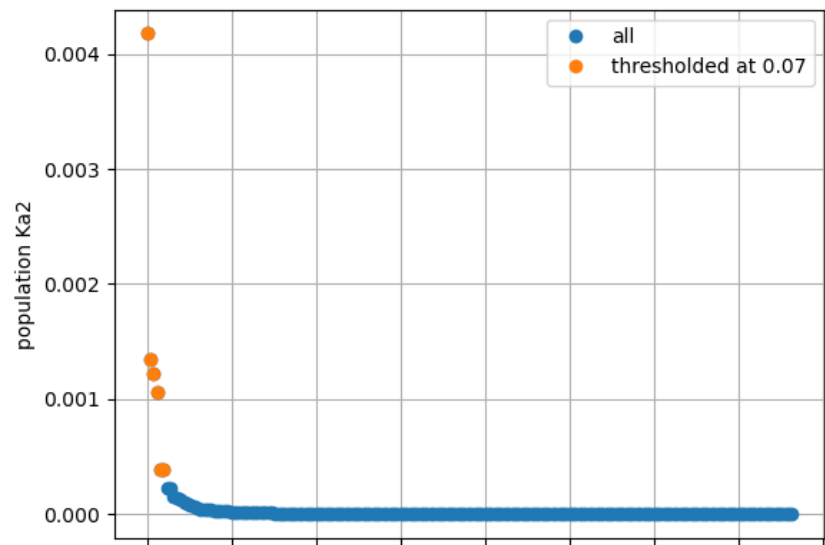
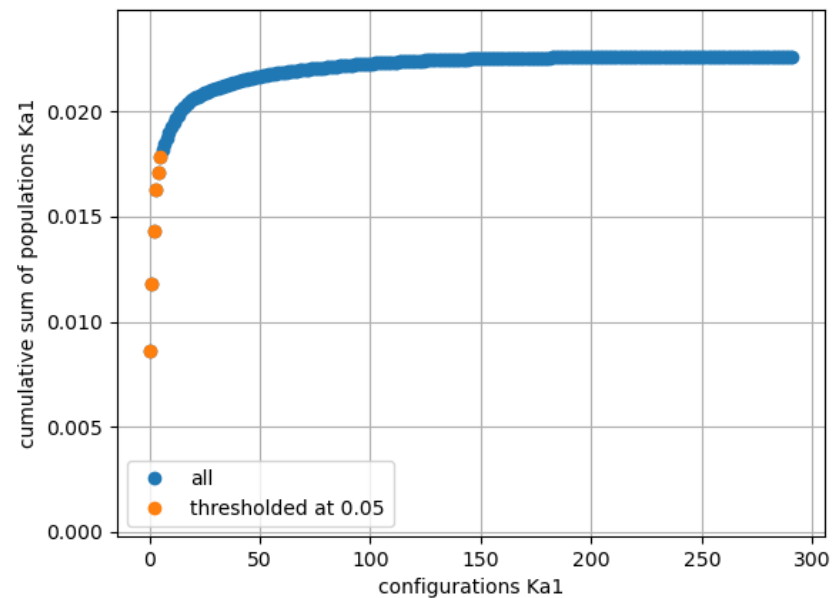
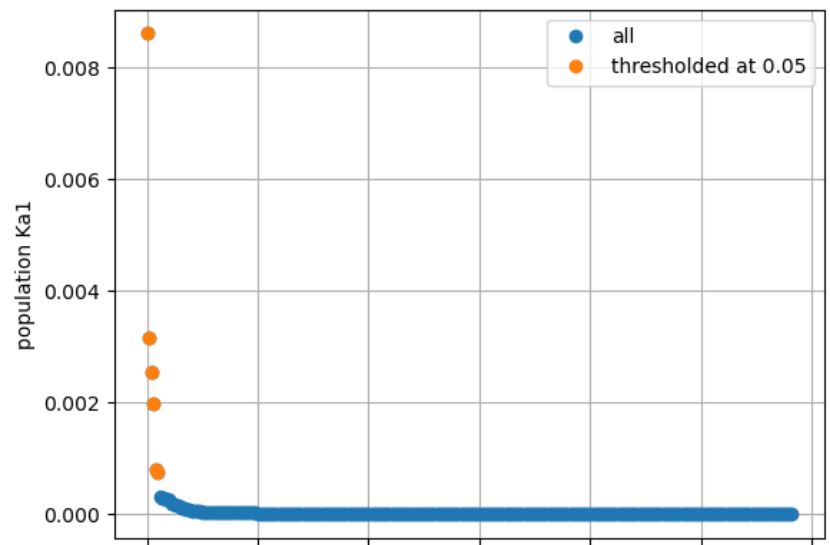


Effective enhancement due PAA cascades = 1.97

Temporal Evolution of Population for 2p5 Configurations



Relativistic (spin-orbit) calculations with XATOM



Relativistic
(spin-orbit)
calculation,

1340
configurations
in total,

267
configurations
that result in K-
a1
fluorescence,

167
configurations
that result in K-
a2

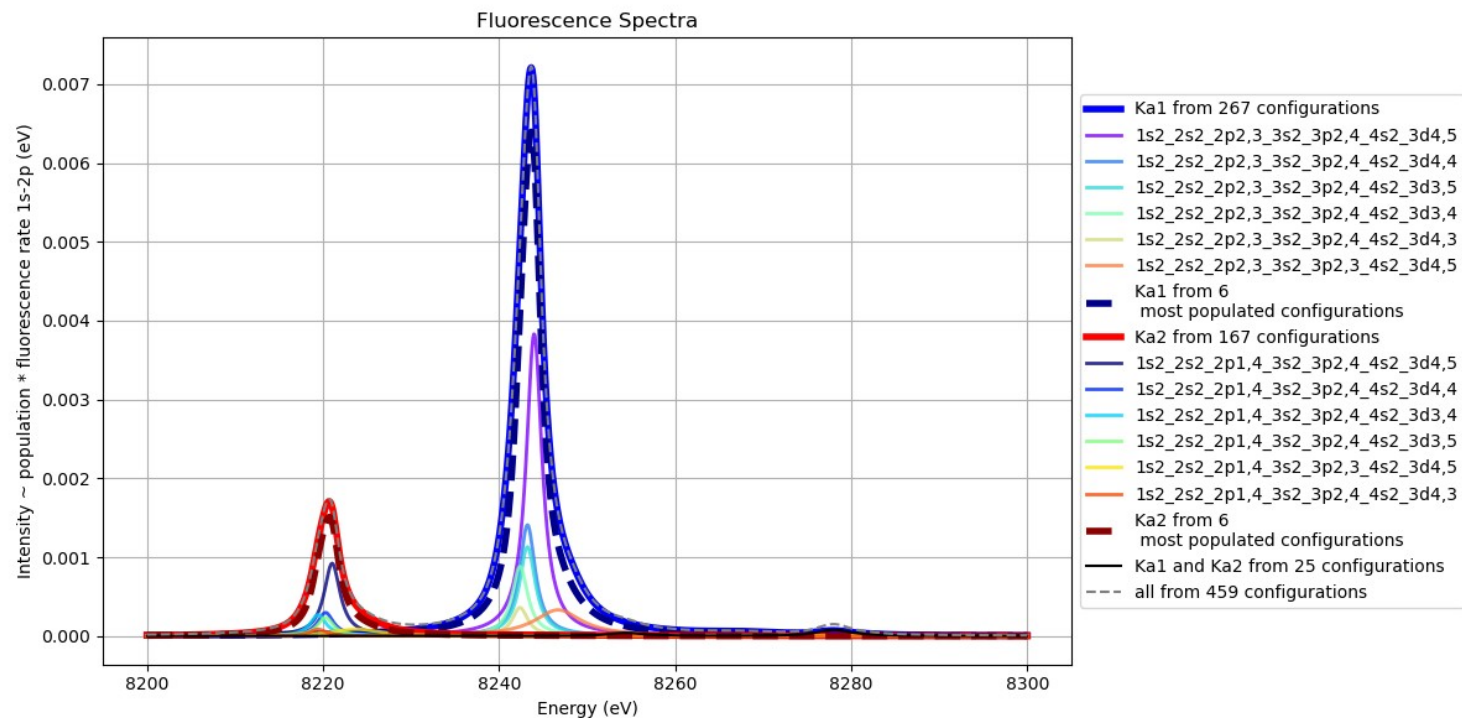
25
configurations
that result in
both K-a1 and 2

Relativistic (spin-orbit) calculation

	width_2p_eV	energy_Ka1_eV	rate_Ka1_eV	width_1s_Ka1_eV
1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,5	0.634635	8243.98	0.445235	1.538088
1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,4	0.606980	8243.22	0.445129	1.544676
1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,5	0.634779	8243.22	0.445129	1.544679
1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,4	0.602928	8242.37	0.445053	1.552652
1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,3	0.572228	8242.37	0.445053	1.552649
1s2_2s2_2p2,3_3s2_3p2,3_4s2_3d4,5	2.632492	8246.69	0.445916	3.852452

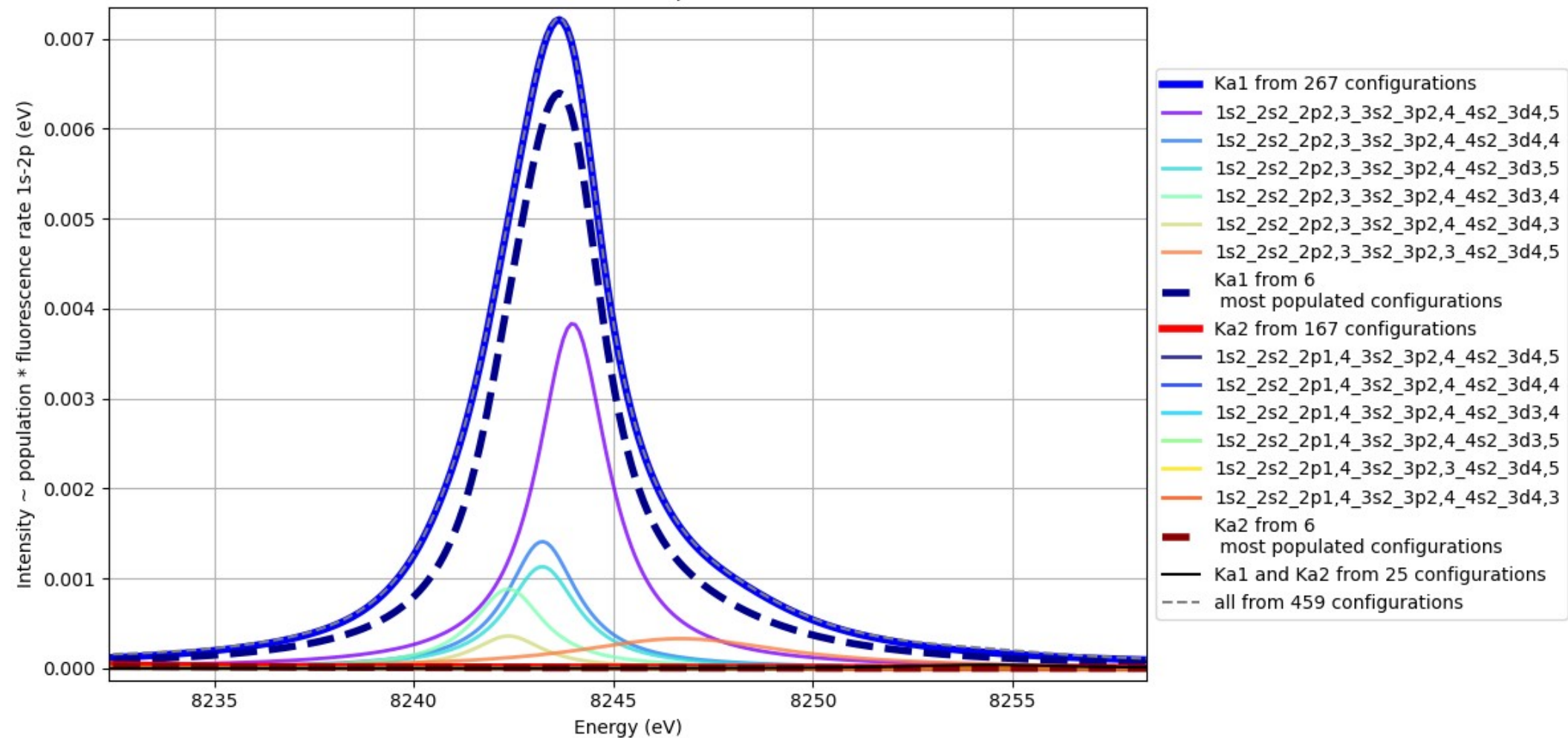
	width_2p_eV	energy_Ka2_eV	rate_Ka2_eV	width_1s_Ka2_eV
1s2_2s2_2p1,4_3s2_3p2,4_4s2_3d4,5	0.665827	8221.03	0.220765	1.538088
1s2_2s2_2p1,4_3s2_3p2,4_4s2_3d4,4	0.655903	8220.28	0.220712	1.544676
1s2_2s2_2p1,4_3s2_3p2,4_4s2_3d3,4	0.609448	8219.44	0.220674	1.552652
1s2_2s2_2p1,4_3s2_3p2,4_4s2_3d3,5	0.609211	8220.28	0.220712	1.544679
1s2_2s2_2p1,4_3s2_3p2,3_4s2_3d4,5	2.723615	8223.70	0.221099	3.852452
1s2_2s2_2p1,4_3s2_3p2,4_4s2_3d4,3	0.651794	8219.44	0.220674	1.552649

Figure 283

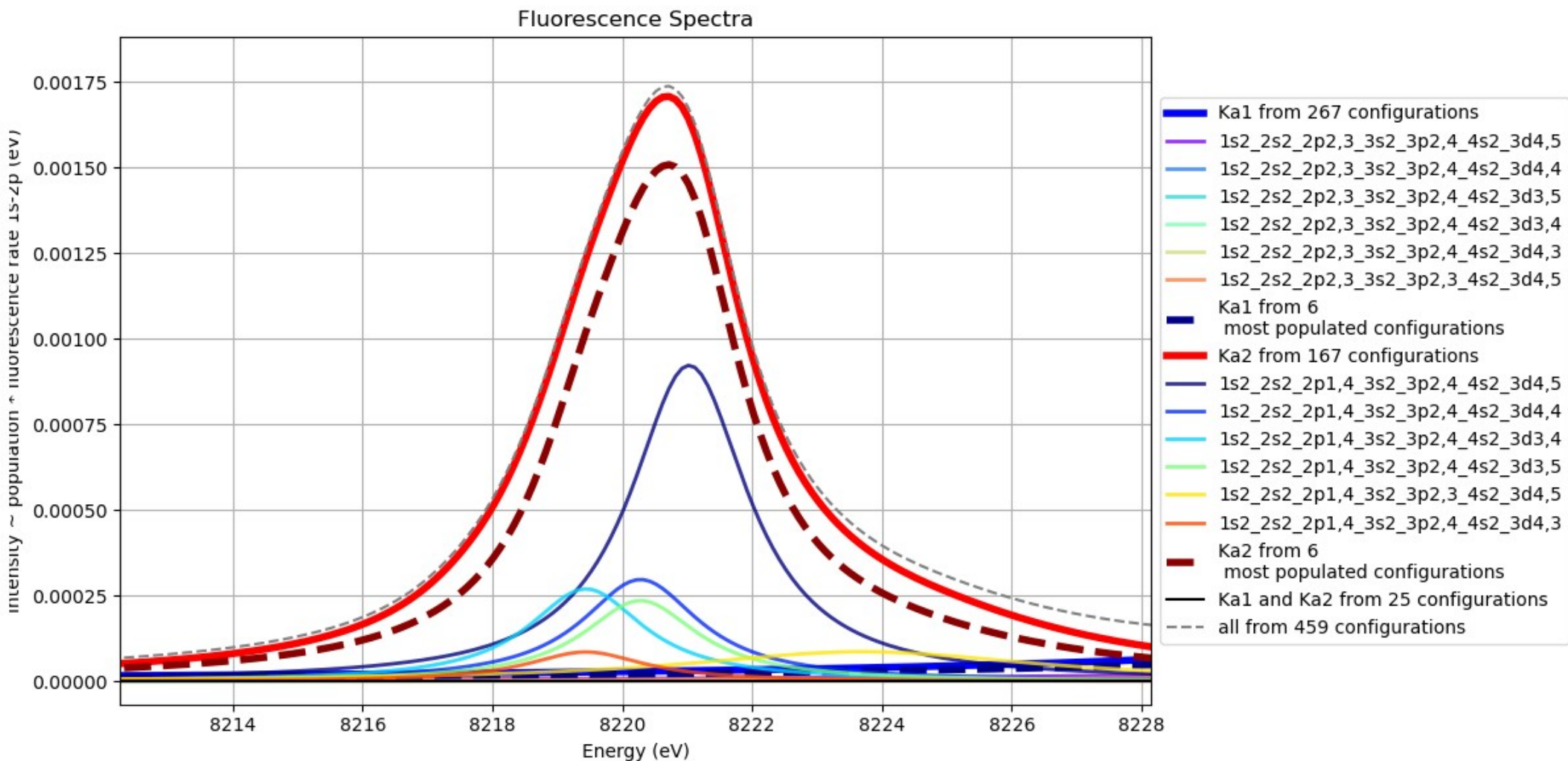


Relativistic (spin-orbit) calculation

Fluorescence Spectra



Relativistic (spin-orbit) calculation



Relativistic (spin-orbit) calculation, effective pathways

Pathways to 1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,4:

Pathways to 1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,3:

Pathways to 1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,5:

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 64.63 fs, br=0.26

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,5

pathway branching ratio = 0.2614

Pathways to 1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,4:

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 35.11 fs, br=0.48

1s2_2s1_2p2,4_3s2_3p2,4_4s2_3d4,5

| A, 0.42 fs, br=0.19

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,4

pathway branching ratio = 0.0927

Pathways to 1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,5:

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 35.11 fs, br=0.48

1s2_2s1_2p2,4_3s2_3p2,4_4s2_3d4,5

| A, 0.50 fs, br=0.16

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,5

pathway branching ratio = 0.0773

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 35.11 fs, br=0.48

1s2_2s1_2p2,4_3s2_3p2,4_4s2_3d4,5

| A, 0.45 fs, br=0.18

1s2_2s2_2p2,3_3s2_3p2,3_4s2_3d4,5

| A, 0.64 fs, br=0.38

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,4

pathway branching ratio = 0.0329

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 35.11 fs, br=0.48

1s2_2s1_2p2,4_3s2_3p2,4_4s2_3d4,5

| A, 1.02 fs, br=0.08

1s2_2s2_2p2,3_3s2_3p1,4_4s2_3d4,5

| A, 0.39 fs, br=0.52

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,4

pathway branching ratio = 0.0199

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 64.63 fs, br=0.26

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,5

| A, 5.01 fs, br=0.20

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d3,4

| P, 64.70 fs, br=0.26

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,4

pathway branching ratio = 0.0132

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 127.86 fs, br=0.13

1s2_2s2_2p1,4_3s2_3p2,4_4s2_3d4,5

| A, 3.78 fs, br=0.25

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d3,4

| P, 64.70 fs, br=0.26

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,4

pathway branching ratio = 0.0085

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 505.76 fs, br=0.03

1s2_2s2_2p2,4_3s2_3p2,3_4s2_3d4,5

| A, 0.42 fs, br=0.55

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d3,4

| P, 64.70 fs, br=0.26

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d3,4

pathway branching ratio = 0.0048

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 35.11 fs, br=0.48

1s2_2s1_2p2,4_3s2_3p2,4_4s2_3d4,5

| A, 0.45 fs, br=0.18

1s2_2s2_2p2,3_3s2_3p2,3_4s2_3d4,5

| A, 1.04 fs, br=0.24

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,3

pathway branching ratio = 0.0204

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 64.63 fs, br=0.26

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,5

| A, 9.14 fs, br=0.11

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,3

| P, 64.70 fs, br=0.26

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,3

pathway branching ratio = 0.0073

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 505.76 fs, br=0.03

1s2_2s2_2p2,4_3s2_3p2,3_4s2_3d4,5

| A, 0.75 fs, br=0.31

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,3

| P, 64.70 fs, br=0.26

1s2_2s2_2p2,3_3s2_3p2,4_4s2_3d4,3

pathway branching ratio = 0.0027

Pathways to 1s2_2s2_2p2,3_3s2_3p2,3_4s2_3d4,5:

1s2_2s2_2p2,4_3s2_3p2,4_4s2_3d4,5

| P, 35.11 fs, br=0.48

1s2_2s1_2p2,4_3s2_3p2,4_4s2_3d4,5

| A, 0.45 fs, br=0.18

1s2_2s2_2p2,3_3s2_3p2,3_4s2_3d4,5

pathway branching ratio = 0.0857

Relativistic (spin-orbit) calculation, effective enhancement due PAA cascades = 2.26

Temporal Evolution of Population for Ka1 Configurations

